DJBILBOX BEATS / HUMPIRE

Complete User Manual

Installation, license and detailed guide for every plugin

Version 1.0 - Windows (VST3) and macOS (VST3 / Audio Unit) - Royalty-free license

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Welcome

Thank you for choosing **DJBILBOX BEATS / HUMPIRE**. This manual covers installation, licensing and the detailed use of every plugin in the collection: the DJBILBOX BEATS synthesizers, the **HUMPIRE LOADOUT** studio effects suite, the **HUMPIRE ORBIT** vocal effects and the **HUMPIRE CHAABI ORCHESTRA** sampler.

Every plugin is designed and programmed in Toulouse, France, by DJBILBOX BEATS, beatmaker and developer. All audio algorithms, interfaces and presets are original creations.

System requirements

Operating system	Windows 10 or Windows 11, 64-bit
Format	VST3 64-bit
Tested host	FL Studio 21 and later (any VST3 host should work: Ableton Live, Cubase, Studio One, Reaper, Bitwig...)
CPU	x64 processor (Intel Core i5 / AMD Ryzen 5 or better recommended)
Memory	4 GB minimum, 8 GB recommended
Display	1280 × 720 minimum; interfaces are resizable
Disk space	About 300 MB for the full collection (excluding your own samples)

Installation

1. Close your music software (FL Studio or any other host).
2. Unzip the downloaded archive and run **DJBILBOX_BEATS_ALL_PLUGINS_Setup.exe**.
3. Accept the Windows administrator prompt. If Windows SmartScreen shows “Windows protected your PC”, click **More info**, then **Run anyway**.
4. Read and accept the license agreement.
5. Choose **Full installation**, or untick the plugins you do not want.
6. Click **Install**. The plugins are copied to the standard Windows VST3 folder: `C:\Program Files\Common Files\VST3`
7. This manual and the license agreement are installed in `C:\Program Files\DJBILBOX BEATS\Documentation` and can be opened from the Start menu (**DJBILBOX BEATS** folder).

First use in FL Studio

1. Open FL Studio, then **Options > Manage plugins**.
2. Make sure `C:\Program Files\Common Files\VST3` is listed among the plugin search paths (it is by default).
3. Click **Find installed plugins** (or **Start scan**). The plugins appear under the vendor **DJBILBOX BEATS** or **HUMPIRE by Djbilbox BEATS**.
4. Click the star (favorite) next to each plugin to find it quickly in the **Plugin database** and the browser.
5. **Synthesizers and sampler**: add them to the **Channel rack** (**+** button) and play them from a MIDI keyboard or the Piano roll.
6. **Effects**: open the **Mixer**, select a track, click an empty slot in the effects chain and choose the plugin.

> Tip: in FL Studio, right-click any plugin control and choose **Link to controller** to drive it from your MIDI controller, or **Create automation clip** to automate it.

Common gestures

Gesture	Action
Drag a knob vertically	Change the value
Shift + drag	Fine adjustment
Double-click	Reset to the default value
Right-click (on supported plugins)	Type a numeric value
Mouse wheel	Small steps
Bottom-right corner of the window	Resize the interface (on supported plugins)

Presets

Each plugin ships with factory presets organized by genre or instrument. Use the **<** and **>** arrows or the preset menu in the top bar. Your own presets are saved in your user folder (`Documents` or `AppData` , depending on the plugin, see the plugin's chapter) and are never overwritten by updates.

Uninstalling

Open **Windows Settings > Apps > Installed apps**, find **DJBILBOX BEATS - All Plugins** and click **Uninstall**. Your own presets and samples are kept.

License summary

The sounds and music you create with these plugins belong to you and are **royalty-free**: you may use them commercially (streaming, sales, placements, sync licensing, beat sales and leases) with no royalties and no mandatory credit. The license is personal and allows installation on two computers you own. Redistributing the plugins, their presets or their factory sounds on their own is prohibited. The full text is in **EULA_DJBILBOX_BEATS.txt**.

Support

Before contacting support, check the FAQ section of the plugin concerned. For any question: **djbilboxbeats.com**. Please include the plugin name, your Windows version, your host software and its version, and a precise description of the issue.

ARENA BASS

Stadium-size bass synthesizer: 808 subs, slap, reese, acid, horns and kicks from eight synthesis engines.

Type	Synthesizer (VST3 instrument)
Format	VST3 64-bit (Windows)
Developer	DJBILBOX BEATS
Version	1.0
Factory presets	70
Latency	0 samples

Overview

ARENA BASS is a bass synthesizer by Humpire / DJBILBOX BEATS. It uses no samples. Every sound is generated in real time by one of eight engines: a sine sub with a pitch “knock”, a plucked-string slap bass, a Moog Bass engine (two saws, a square and a sub octave), a detuned Stadium Reese, a two-operator FM Growl, an Acid Rubber voice with a self-closing resonant filter, a Stadium Horns brass section and an Arena Kick for pitched kicks and stomps.

Five character controls (TONE, BODY, BRIGHT, WIDTH, TEXTURE) reshape each engine. The sound then goes through a multimode filter with its own envelope, an amplitude envelope, an LFO and a master effects chain: distortion, chorus, ping-pong delay and reverb. The character is big, punchy and direct, made for sport-arena energy and hard-hitting low end.

ARENA BASS targets Trap, Drill and Hip-Hop 808s, Afrotrap and Afro bass lines, G-Funk slides, Deep House and acid bass, plus horn stabs and anthem hooks. Bass presets load in Mono mode for clean, one-note-at-a-time lines. The Stadium Horns bank plays chords in Poly mode with up to 16 voices.

Quick start in FL Studio

1. In the **Channel rack**, click **+** and choose **ARENA_BASS** (vendor **DJBILBOX BEATS**). You can also drag it from the **Plugin database > Generators** browser.
2. The plugin opens on **ARENA BASS - Init Patch**: the Arena Sub engine in Poly mode.
3. Click the **folder icon** at the left of the preset display. Pick a bank, then a preset. You can also step through all 70 presets with the **<** and **>** arrows.
4. Play from your MIDI keyboard or draw notes in the **Piano roll**. Bass lines sit best between MIDI notes 24 and 48 (C2 to C4 in FL Studio). Presets with PITCH at -12 sound one octave lower than the key you play.
5. Shape the sound with the three main controls:
   - **TONE**: the main character of the engine (knock, pluck hardness, sub level, FM ratio...).
   - **FILTER**: the cutoff of the filter. Turn it down for a darker, rounder bass.
   - **GLIDE**: slide time between notes. Leave it at zero for straight notes.
6. Adjust **VOLUME** to set the level, then click the **disk icon** to save your own version.

Interface

The window opens at 1000 × 746 pixels. Drag the bottom-right corner to resize it from 50% to 200%. The proportions stay fixed.

Preset display (top center). From left to right: the folder icon (preset menu), the **<** arrow, the current preset name, the **>** arrow and the disk icon (save). The arrows wrap around from the last preset to the first.

EXPANSION cassette (top left). Clicking the cassette opens an information message about upcoming expansion packs. No expansion content is included with version 1.0.

Logo (top right). The DJBILBOXBEATS name and the DB logo.

Center. The camel artwork, the ARENA BASS title and “By Humpire”. The dotted grille at the left of the artwork is decorative.

Character panel (top right, slatted panel). Two rows of small knobs:

- Row 1: **TONE, BODY, BRIGHT, WIDTH, TEXTURE**. The five character controls of the selected engine.
- Row 2: **RES** (filter resonance), **ENV** (filter envelope depth), **DISTORTION** and **DELAY** (delay level).

LFO section (left). RATE and **AMOUNT** set the LFO speed and depth. **PITCH**, also under this header, is the global transposition (-12 to +12 semitones). It is not an LFO control.

Output and effects (right). VOLUME, **REVERB** and **CHORUS**. The header above this row reads “ADSR”, but these knobs are the output level and two effects. The amplitude envelope is the slider bank described below.

Center row. From left to right:

- **ATTACK, DECAY, SUSTAIN, RELEASE**: the amplitude envelope sliders.
- **GLIDE** and **FILTER** knobs.
- **A, D, S, R**: the filter envelope sliders. **VELOCITY**: velocity sensitivity of the level.
- **TIME** and **FEEDBACK**: delay time and delay repeats.

Menu row. Five drop-down menus, from left to right: **Waveform** (engine), **LFO Source** (LFO shape), **Destination** (LFO target), **Filter Type** and **Mono/Poly**. Each menu shows its current value.

Wheels (bottom left). Pitch Bend springs back to the center when you release it. **Modulation** stays where you leave it.

On-screen keyboard (bottom). Click a key to play it. Velocity follows the click position: click near the bottom of a key for a louder note. The keyboard starts at the lowest MIDI note (labeled C-2, which is C0 in FL Studio) and does not scroll. It covers MIDI notes 0 to 31, so the left half is below the audible range. Use a MIDI controller or the Piano roll for normal playing.

Mouse gestures. Drag a knob up or right to increase it. Drag sliders and wheels vertically. Double-click any knob or slider to return it to its default value. The mouse wheel also adjusts controls. A value bubble appears while you drag.

Parameters

Signal flow: engine, then filter, then amplitude envelope (per voice). The voices are summed, then processed by Distortion, Chorus, Delay and Reverb. A DC blocker and a soft limiter protect the output.

ARENA BASS exposes 35 automatable parameters. The host name is shown in parentheses when it differs from the panel label.

Parameter	Range	Default	Function
Waveform (menu)	Arena Sub, Slap Bass, Moog Bass, Stadium Reese, FM Growl, Acid Rubber, Stadium Horns, Arena Kick	Arena Sub	Selects the synthesis engine.
TONE (Tone)	0 to 1	0.5	Character control 1. Its effect depends on the engine (see the table below).
BODY (Body)	0 to 1	0.5	Character control 2. Engine-dependent.
BRIGHT (Brightness)	0 to 1	0.5	Character control 3. Engine-dependent.
WIDTH (Width)	0 to 1	0.3	Character control 4. Engine-dependent (detune, drive, wobble speed...). It is not a stereo width control.
TEXTURE (Texture)	0 to 1	0.2	Character control 5. Engine-dependent (saturation, noise, feedback...).
FILTER (Filter)	40 to 18,000 Hz	9,000 Hz	Filter cutoff. The cutoff follows the keyboard by 25% around middle C.
RES (Resonance)	0 to 1	0.15	Filter resonance.
Filter Type (menu)	Low Pass, High Pass, Band Pass	Low Pass	Filter mode.
ENV (Filter Env)	0 to 1	0	Filter envelope depth. At 1 the envelope opens the cutoff up to 5 octaves. At 0 the filter envelope has no effect.
A (Filter Attack)	0.002 to 3 s	0.002 s	Filter envelope attack time.
D (Filter Decay)	0.02 to 4 s	0.3 s	Filter envelope decay time.
S (Filter Sustain)	0 to 1	0	Filter envelope sustain level.
R (Filter Release)	0.02 to 4 s	0.3 s	Filter envelope release time.
VELOCITY (Velocity)	0 to 1	0.5	How much note velocity changes the level. 0: all notes at the same level. 1: full velocity response.
ATTACK (Attack)	0.001 to 5 s	0.01 s	Amplitude envelope attack time.
DECAY (Decay)	0.01 to 5 s	0.4 s	Amplitude envelope decay time.
SUSTAIN (Sustain)	0 to 1	0.7	Amplitude envelope sustain level.

Parameter	Range	Default	Function
RELEASE (Release)	0.02 to 8 s	0.4 s	Amplitude envelope release time.
VOLUME (Volume)	0 to 1	0.7	Level of each voice, before the effects.
GLIDE (Glide)	0 to 2	0	Portamento. The value is roughly the slide time in seconds. Each new note slides from the pitch of the previous note, in Mono and in Poly mode. 0 turns glide off.
PITCH (Pitch)	-12 to +12 semitones	0	Global transposition. The control is continuous: use whole values to stay in tune.
RATE (LFO Rate)	0.05 to 20 Hz	4 Hz	LFO speed. Not tempo-synced.
AMOUNT (LFO Amount)	0 to 1	0	LFO depth. The mod wheel adds to this value.
LFO Source (menu)	Sine, Triangle, Saw, Square, Sample&Hold	Sine	LFO shape. The LFO restarts at every new note. Sample&Hold picks a new random value once per cycle.
Destination (menu)	Pitch, Filter, Amp, Timbre	Pitch	LFO target. Pitch: up to +/-1 semitone (vibrato). Filter: up to +/-3 octaves (wobble). Amp: up to 50% level dip (tremolo). Timbre: moves all five character controls by up to +/-0.5.
Mono/Poly (menu)	Poly, Mono	Poly	Poly: up to 16 notes at once. Mono: one note at a time, last-note priority.
DISTORTION (Distortion)	0 to 1	0	Saturation on the summed output, with automatic level compensation.
CHORUS (Chorus)	0 to 1	0	Stereo chorus mix (slow, fixed rate).
DELAY (Delay)	0 to 1	0	Level of the ping-pong delay.
TIME (Delay Time)	0.05 to 1.0 s	0.35 s	Delay time in seconds. Not tempo-synced: for a quarter note, use 60 / BPM (0.5 s at 120 BPM); halve it for an eighth note.
FEEDBACK (Feedback)	0 to 0.9	0.35	Number of delay repeats (0 to 90% feedback).

Parameter	Range	Default	Function
REVERB (Reverb)	0 to 1	0.25	Reverb amount. Higher values also enlarge the room.
Pitch Bend wheel (Pitch Bend)	-1 to +1	0	On-screen pitch bend, +/-2 semitones. Adds to MIDI pitch bend.
Modulation wheel (Mod Wheel)	0 to 1	0	On-screen mod wheel. Adds up to +0.8 to the LFO AMOUNT.

Character controls by engine

Engine	TONE	BODY	BRIGHT	WIDTH	TEXTURE
Arena Sub	Pitch “knock” at the attack	Adds the octave harmonic	Attack click	No effect	Saturation
Slap Bass	Pluck hardness	String sustain	Brighter, longer highs	Low body resonance (around 95 Hz)	Slap click
Moog Bass	Sub-octave level	Output weight	Built-in low-pass opening	Detune of the second saw	Blend from saws to square
Stadium Reese	Sub-octave level	Detune amount	Built-in low-pass opening	Saturation drive	Noise
FM Growl	Modulator ratio (0.5 to 3.5)	No effect	FM depth (growl intensity)	Speed of the built-in wobble (0.5 to 12.5 Hz)	Modulator feedback (grit)
Acid Rubber	Saw to pulse blend	Length of the built-in filter sweep	Depth of the built-in filter sweep	No effect	Saturation
Stadium Horns	Swell time of the built-in filter	Output weight	How far the filter opens	Detune of the three saws	Breath noise
Arena Kick	Pitch-drop depth	Decay length	Click	No effect	Saturation

Factory presets

ARENA BASS includes 70 factory presets in 7 banks of 10. All banks except Stadium Horns load in Mono mode. In the Stadium Horns bank, only Lead Horn is in Mono mode (with a short glide).

Arena Sub: Arena 808, Stadium Rumble, Long Slide Sub, Locker Room Sub, Hard Knock Bass, Short Pump, Smooth Court, Pitch Drop Dunk, Dirty Overtime, Scoreboard Sub.

Slap & Pluck Bass: Slap Bass, Pick Bass, Finger Funk, Muted Pocket, Upright Arena, Pop Slap, Deep Pluck, Chorus Bass, Dub Pluck, Walking Court.

Moog Bass: Arena Moog, Fat Moog, Rubber Moog, Sub Moog, Slide Moog, Pump Bass, Octave Moog, Hot Seat, Bass Stab, Warm Anthem.

Reese & Growl: Stadium Reese, Dirty Reese, Wobble Arena, Industrial Reese, Sub Reese, FM Growl, Talking Bass, Robot Growl, Metal Growl, Slow Growl.

Acid & Rubber: Acid Rubber, Squelch, Rubber Band, Acid Slide, Pulse Acid, Dirty Acid, Open Acid, Low Rubber, Pump Squelch, Buzzer Beater.

Stadium Horns: Stadium Horns, Fanfare, Brass Stab, Victory Horns, Soft Flugel, Slow Swell, Hit Section, Low Horns, Lead Horn, Anthem.

Kicks & Stomps: Arena Kick, Stomp, Long Boom, Punch Kick, Sub Kick, Crowd Stomp, Dirty Kick, Tom Fill, Drop Impact, Laser Kick.

Loading a preset first resets every parameter to its default value, then applies the preset. The two wheels are not changed.

User presets. Click the disk icon to save the current sound as a `.abpreset` file. The default folder is `Documents\DJBILBOX BEATS\ARENA_BASS\Presets`. Files in this folder appear in the **User** submenu of the preset menu. Use **Open a .abpreset file...** in the same menu to load a file from another location. The preset display shows the file name.

The host also sees the 70 factory presets as a program list. Your project saves the full state of the plugin, including the preset name.

Moog is a trademark of its owner. It is used here only to describe a sound style. DJBILBOX BEATS is not affiliated with the trademark owner.

Production tips

1. **Trap and Drill 808s.** Start from Arena 808 or Long Slide Sub. Raise TONE for a harder knock on short notes. Raise TEXTURE or add a little DISTORTION so the 808 stays audible on phone and laptop speakers.
2. **808 slides.** With GLIDE above zero, every new note slides from the previous one, even if the notes do not overlap. For slides on selected notes only, keep GLIDE at 0 and automate it up just before the notes that should slide.
3. **G-Funk bass.** Use the Moog Bass bank (Slide Moog, Fat Moog). Set GLIDE around 0.1 to 0.2 and play legato lines. Raise ENV and shorten the filter D slider for a rounder “pluck” at the start of each note.
4. **Afrotrap and Afro bass.** The Slap & Pluck bank (Muted Pocket, Finger Funk, Pick Bass) gives bouncy, percussive lines. These presets are quieter by nature, so raise VOLUME. Use VELOCITY around 0.6 to 0.8 so accents in your Piano roll come through.
5. **Deep House and acid.** Load Arena Moog or a preset from the Acid & Rubber bank. Automate FILTER over 8 or 16 bars. For movement, set Destination to Filter with a Square or Sample&Hold LFO, as in Pump Squelch.
6. **Reese and wobble.** On the Stadium Reese engine, BODY sets the detune and WIDTH drives the saturation. Set Destination to Filter, LFO Source to Sine and RATE between 1 and 4 Hz for a wobble (see Wobble Arena).
7. **Horn hooks and anthems.** The Stadium Horns bank plays chords in Poly mode. Raise ATTACK for swells (Slow Swell) or shorten DECAY and SUSTAIN for stabs (Hit Section). Add REVERB and CHORUS for a stadium-size sound.
8. **Kicks and stomps.** Load the Kicks & Stomps presets in a second ARENA BASS instance, so the kick and the 808 keep separate envelopes. Play kicks around MIDI notes 28 to 40 (E2 to E3 in FL Studio). TONE sets the pitch drop, BODY the length and BRIGHT the click. Tom Fill is transposed one octave up for fills.

MIDI, automation and performance

Notes and channels. ARENA BASS responds to all 128 MIDI notes on any MIDI channel. Velocity controls the level according to the VELOCITY slider.

Polyphony. Up to 16 voices in Poly mode. When you play a 17th note, a voice is taken over and cut immediately. In Mono mode, each new note stops the previous one (last-note priority). Releasing the newest note does not bring back an earlier note that you still hold.

Pitch bend. MIDI pitch bend covers +/-2 semitones. The on-screen Pitch Bend wheel adds another +/-2 semitones.

Mod wheel (CC1). CC1 raises the LFO depth, like the on-screen Modulation wheel. When both are used, the higher value wins. CC1 acts on the notes that are sounding when the message arrives, so move the wheel while you play. For repeatable results in a song, automate the on-screen **Mod Wheel** parameter instead.

Other MIDI. The sustain pedal (CC64) is supported. All Notes Off and All Sound Off messages stop the sound. Aftertouch is ignored.

Automation in FL Studio. All 35 parameters can be automated. Move a control in the plugin, then use **Tools > Last tweaked > Create automation clip**. Use **Tools > Last tweaked > Link to controller** to assign a hardware knob. You can also pick parameters from the plugin wrapper menu (**Browse parameters**). Level, pitch, bend, filter, distortion and delay changes are smoothed to avoid clicks.

Latency and tail. The plugin adds no latency (0 samples), so no delay compensation is needed. It reports a 12-second tail to the host so reverb and delay trails are not cut when rendering.

CPU. ARENA BASS is lightweight. Effects set to zero are bypassed and cost no CPU. Mono mode and short release times keep the voice count low. The plugin works at 44.1, 48, 88.2 and 96 kHz.

FAQ

ARENA BASS does not appear in FL Studio. Open **Options > Manage plugins**, check that `C:\Program Files\Common Files\VST3` is in the search paths, then click **Find installed plugins**. The plugin is listed as **ARENA_BASS** under **DJBILBOX BEATS**. It is an instrument: load it in the Channel rack, not in a Mixer insert slot.

I hear nothing, or almost nothing, when I click the on-screen keyboard. The on-screen keyboard covers the lowest MIDI notes (0 to 31), and most of them are below what speakers can reproduce. Play from a MIDI keyboard or the Piano roll around C2 to C5 (FL Studio naming). Also check VOLUME, and check that FILTER is not closed with Filter Type set to Low Pass.

My 808 is not perfectly centered. Each note gets a slight random stereo offset. For a strictly centered sub, reduce the stereo separation of the mixer track in FL Studio, or insert a mono utility after ARENA BASS. If the bass is also weak on small speakers, add harmonics with TEXTURE, BODY or DISTORTION.

Every note slides into the next one. GLIDE is above zero. Glide applies to every new note. Set GLIDE to 0, or automate it only where you want slides.

My chord only plays one note. The Mono/Poly menu is set to Mono, which is the default for bass presets. Switch it to Poly.

CHINTZWAZ

Retro floral synthesizer: sitar, flower-power keys, tape strings, flutes, kalimba, pads and psychedelic organ.

Type	Synthesizer (VST3 instrument)
Format	VST3 64-bit (Windows)
Developer	DJBILBOX BEATS
Version	1.0
Factory presets	70
Latency	0 samples

Overview

CHINTZWAZ is a retro, bohemian synthesizer by Humpire / DJBILBOX BEATS. It uses no samples. Eight engines recreate the colors of the psychedelic era: a plucked Sitar Bloom with jawari buzz, a Flower EP electric piano with velocity “bark”, a Mellotron tape-string voice with flutter and hiss, a breathy Hippie Flute with vibrato, a Kalimba, a slowly breathing Daisy Pad, a round Bloom Bass and a Psyche Organ with drawbars and tremolo.

Five character controls (TONE, BODY, BRIGHT, WIDTH, TEXTURE) reshape each engine. The sound then goes through a multimode filter with its own envelope, an amplitude envelope, an LFO and a master effects chain: distortion, chorus, ping-pong delay and reverb. The character is warm, colorful and slightly worn, like an old record or a tape loop.

CHINTZWAZ fits Lo-fi Hip-Hop and Boom Bap sample-style loops, psychedelic Soul and Funk, Oriental and Rai melodies (sitar, flute), Afrobeats and Amapiano (kalimba, marimba), Deep House keys and organs, and melodic Trap and R&B. It plays up to 16 notes at once. Flute leads and bass presets load in Mono mode.

Quick start in FL Studio

1. In the **Channel rack**, click **+** and choose **CHINTZWAZ** (vendor **DJBILBOX BEATS**). You can also drag it from the **Plugin database > Generators** browser.
2. The plugin opens on **CHINTZWAZ - Init Patch**: the Sitar Bloom engine in Poly mode.
3. Click the **folder icon** at the left of the preset display. Pick a bank, then a preset. You can also step through all 70 presets with the **<** and **>** arrows.
4. Play from your MIDI keyboard or draw notes in the **Piano roll**. Keys, pads and plucks sit well between C4 and C7 in FL Studio. Bass presets sit between C2 and C4.
5. Shape the sound with the three main controls:
   - **TONE**: the main character of the engine (pluck hardness, tape color, breath pitch, organ sub drawbar...).
   - **FILTER**: the cutoff of the filter. Turn it down for a darker, vintage tone.
   - **WIDTH**: jawari buzz, tape flutter, flute vibrato or organ tremolo, depending on the engine.
6. Adjust **VOLUME** to set the level, then click the **disk icon** to save your own version.

Interface

The window opens at 1000 × 746 pixels on a pink brushed-metal faceplate with floral wallpaper. Drag the bottom-right corner to resize it from 50% to 200%. The proportions stay fixed.

Preset display (top center). From left to right: the folder icon (preset menu), the **<** arrow, the current preset name, the **>** arrow and the disk icon (save). The arrows wrap around from the last preset to the first.

EXPANSION cassette (top left). Clicking the cassette opens an information message about upcoming expansion packs. No expansion content is included with version 1.0.

Logo (top right). The DB logo and the DJBILBOXBEATS name.

Left speaker grille. Four small knobs stacked vertically: **TONE, BODY, BRIGHT, WIDTH**.

Right speaker grille. Three small knobs: **TEXTURE** (the fifth character control), **RES** (filter resonance) and **ENV** (filter envelope depth).

Center. The camel artwork, the CHINTZWAZ title and “By Humpire”. The **GLIDE** knob sits at the left of the title and the **FILTER** knob at the right.

LFO section (left). RATE and **AMOUNT** set the LFO speed and depth. **PITCH**, also under this header, is the global transposition (-12 to +12 semitones). It is not an LFO control.

Amplitude envelope (lower left). Four sliders: **ATTACK, DECAY, SUSTAIN, RELEASE**.

ADSR+VELOCITY section (upper right). Four sliders: **ATTACK, DECAY, SUSTAIN** of the filter envelope, and **VELOCITY** (velocity sensitivity of the level). The filter envelope has no separate release slider: its release time follows its DECAY slider. The **VOLUME** knob sits at the right of these sliders.

Effects (right). First row: **REVERB, CHORUS, DISTORTION**. Second row: **DELAY** (delay level), **TIME** and **FEEDBACK**.

Menu row. Five drop-down menus, from left to right: **Waveform** (engine), **LFO Source** (LFO shape), **Destination** (LFO target), **Filter Type** and **Mono/Poly**. Each menu shows its current value.

Wheels (bottom left). PITCH is the pitch bend wheel. It springs back to the center when you release it. **MODULATION** stays where you leave it.

On-screen keyboard (bottom). Click a key to play it. Velocity follows the click position: click near the bottom of a key for a louder note. The keyboard starts at the lowest MIDI note (labeled C-2, which is C0 in FL Studio) and does not scroll. It covers MIDI notes 0 to 31, which are very low for most CHINTZWAZ sounds. Use a MIDI controller or the Piano roll for normal playing.

Mouse gestures. Drag a knob up or right to increase it. Drag sliders and wheels vertically. Double-click any knob or slider to return it to its default value. The mouse wheel also adjusts controls. A value bubble appears while you drag.

Parameters

Signal flow: engine, then filter, then amplitude envelope (per voice). The voices are summed, then processed by Distortion, Chorus, Delay and Reverb. A DC blocker and a soft limiter protect the output.

CHINTZWAZ exposes 34 automatable parameters. The host name is shown in parentheses when it differs from the panel label.

Parameter	Range	Default	Function
Waveform (menu)	Sitar Bloom, Flower EP, Mellotron, Hippie Flute, Kalimba, Daisy Pad, Bloom Bass, Psyche Organ	Sitar Bloom	Selects the synthesis engine.
TONE (Tone)	0 to 1	0.5	Character control 1. Its effect depends on the engine (see the table below).

Parameter	Range	Default	Function
BODY (Body)	0 to 1	0.5	Character control 2. Engine-dependent.
BRIGHT (Brightness)	0 to 1	0.5	Character control 3. Engine-dependent.
WIDTH (Width)	0 to 1	0.3	Character control 4. Engine-dependent (buzz, flutter, vibrato, tremolo...). It is not a stereo width control.
TEXTURE (Texture)	0 to 1	0.2	Character control 5. Engine-dependent (noise, breath, hiss, click...).
FILTER (Filter)	40 to 18,000 Hz	9,000 Hz	Filter cutoff. The cutoff follows the keyboard by 25% around middle C.
RES (Resonance)	0 to 1	0.15	Filter resonance.
Filter Type (menu)	Low Pass, High Pass, Band Pass	Low Pass	Filter mode.
ENV (Filter Env)	0 to 1	0	Filter envelope depth. At 1 the envelope opens the cutoff up to 5 octaves. At 0 the filter envelope has no effect.
ATTACK, filter section (Filter Attack)	0.002 to 3 s	0.002 s	Filter envelope attack time.
DECAY, filter section (Filter Decay)	0.02 to 4 s	0.3 s	Filter envelope decay time. Also sets the filter envelope release time.
SUSTAIN, filter section (Filter Sustain)	0 to 1	0	Filter envelope sustain level.
VELOCITY (Velocity)	0 to 1	0.5	How much note velocity changes the level. 0: all notes at the same level. 1: full velocity response.
ATTACK (Attack)	0.001 to 5 s	0.01 s	Amplitude envelope attack time.
DECAY (Decay)	0.01 to 5 s	0.4 s	Amplitude envelope decay time.
SUSTAIN (Sustain)	0 to 1	0.7	Amplitude envelope sustain level.
RELEASE (Release)	0.02 to 8 s	0.4 s	Amplitude envelope release time.
VOLUME (Volume)	0 to 1	0.7	Level of each voice, before the effects.

Parameter	Range	Default	Function
GLIDE (Glide)	0 to 2	0	Portamento. The value is roughly the slide time in seconds. Each new note slides from the pitch of the previous note, in Mono and in Poly mode. 0 turns glide off.
PITCH (Pitch)	-12 to +12 semitones	0	Global transposition. The control is continuous: use whole values to stay in tune.
RATE (LFO Rate)	0.05 to 20 Hz	4 Hz	LFO speed. Not tempo-synced.
AMOUNT (LFO Amount)	0 to 1	0	LFO depth. The mod wheel adds to this value.
LFO Source (menu)	Sine, Triangle, Saw, Square, Sample&Hold	Sine	LFO shape. The LFO restarts at every new note. Sample&Hold picks a new random value once per cycle.
Destination (menu)	Pitch, Filter, Amp, Timbre	Pitch	LFO target. Pitch: up to +/-1 semitone (vibrato). Filter: up to +/-3 octaves. Amp: up to 50% level dip (tremolo). Timbre: moves all five character controls by up to +/-0.5.
Mono/Poly (menu)	Poly, Mono	Poly	Poly: up to 16 notes at once. Mono: one note at a time, last-note priority.
DISTORTION (Distortion)	0 to 1	0	Saturation on the summed output, with automatic level compensation.
CHORUS (Chorus)	0 to 1	0	Stereo chorus mix (slow, fixed rate).
DELAY (Delay)	0 to 1	0	Level of the ping-pong delay.
TIME (Delay Time)	0.05 to 1.0 s	0.35 s	Delay time in seconds. Not tempo-synced: for a quarter note, use 60 / BPM (0.5 s at 120 BPM); halve it for an eighth note.
FEEDBACK (Feedback)	0 to 0.9	0.35	Number of delay repeats (0 to 90% feedback).
REVERB (Reverb)	0 to 1	0.25	Reverb amount. Higher values also enlarge the room.
PITCH wheel (Pitch Bend)	-1 to +1	0	On-screen pitch bend, +/-2 semitones. Adds to MIDI pitch bend.

Parameter	Range	Default	Function
MODULATION wheel (Mod Wheel)	0 to 1	0	On-screen mod wheel. Adds up to +0.8 to the LFO AMOUNT.

Character controls by engine

Engine	TONE	BODY	BRIGHT	WIDTH	TEXTURE
Sitar Bloom	Pluck hardness	String sustain	Brighter, longer highs	Jawari buzz	Resonant buzz band (900 Hz to 2.5 kHz)
Flower EP	No effect	Decay length	“Bark” (bell and upper partials), stronger at high velocity	No effect	Attack noise
Mellotron	Tape color (formant from 400 Hz to 1.1 kHz)	Formant amount	Built-in low-pass opening	Detune and tape-flutter depth	Flutter speed and tape hiss
Hippie Flute	Pitch of the breath noise	Focus of the breath noise	2nd and 3rd harmonics	Vibrato depth (5.2 Hz, fades in after the attack)	Breath level
Kalimba	Attack click	Decay length	Metallic upper partials	No effect	Box resonance
Daisy Pad	No effect	Sub-octave level	Filter opening (the filter slowly “breathes”)	Detune spread of the five saws	Air noise
Bloom Bass	Triangle to saw blend	Sub-octave level	Depth of the built-in filter sweep	Length of the built-in filter sweep	Saturation
Psyche Organ	Sub-octave drawbar	Octave drawbar	Upper drawbars	Tremolo depth and speed (3 to 7 Hz)	Key click

Factory presets

CHINTZWAZ includes 70 factory presets in 7 banks of 10. Most presets load in Poly mode. These presets load in Mono mode: Violin Tape, Breathy Dawn, Sad Daisy, Flute Echo, Wind In Petals, Whistling Gardener, Vox Garden, and the five bass presets (Bloom Bass, Round Sub, Fuzz Petal, Slide Stem, Pluck Root).

Sitar & Strings: Sitar Bloom, Raga Morning, Patchouli Pluck, Jawari Buzz, Marigold Drone, Soft Tamboura, Hippie Strum, Bazaar Echo, Low Sitar, Glass Sitar.

Flower EP: Flower Power EP, Velvet Daisy, Bark Petal, Summer Haze Keys, Tremolo Bloom, Tape Wurly, Bell Blossom, Low Tide EP, Lazy Sunday, Wah Garden.

Mellotron & Choir: Chintz Mellotron, Strawberry Tape, Soft Choir Tape, Flute Mellotron, Violin Tape, Slow Wobble, Dream Tape Pad, Low Choir Tape, Eight Track, Kaleidoscope.

Hippie Flutes: Hippie Flute, Meadow Pipe, Breathy Dawn, Sad Daisy, High Piccolo, Flute Echo, Wind In Petals, Low Flute, Fog Flute, Whistling Gardener.

Kalimba & Bells: Daisy Kalimba, Music Box Garden, Thumb Piano, Wind Chimes, Glass Petals, Low Kalimba, Echo Garden, Rain On Leaves, Teacup Bell, Marimba Blush.

Daisy Pads: Daisy Pad, Petal Mist, Sundress Strings, Slow Bloom, Lavender Haze, Warm Chintz, Breeze Drift, Whisper Field, Pulse Of Spring, Twilight Garden.

Bloom Bass & Organs: Bloom Bass, Round Sub, Fuzz Petal, Slide Stem, Pluck Root, Psyche Organ, Vox Garden, Soft Flute Stop, Rotary Daisy, Golden Haze Organ.

Loading a preset first resets every parameter to its default value, then applies the preset. The two wheels are not changed.

User presets. Click the disk icon to save the current sound as a `.czpreset` file. The default folder is `Documents\DJBILBOX BEATS\CHINTZWAZ\Presets`. Files in this folder appear in the **User** submenu of the preset menu. Use **Open a .czpreset file…** in the same menu to load a file from another location. The preset display shows the file name.

The host also sees the 70 factory presets as a program list. Your project saves the full state of the plugin, including the preset name.

Mellotron, Vox and Wurlitzer are trademarks of their respective owners. They are used here only to describe sound styles. DJBILBOX BEATS is not affiliated with these trademark owners.

Production tips

1. **Oriental and Rai melodies.** Start from Sitar Bloom or Jawari Buzz. WIDTH sets the jawari buzz and TEXTURE moves the buzzing band. For ney-like flute lines, use Hippie Flute in Mono mode with GLIDE between 0.05 and 0.15, and bend notes with the PITCH wheel (up to 2 semitones).
2. **Lo-fi Hip-Hop and Boom Bap keys.** Load Tape Wurly, Lazy Sunday or Velvet Daisy. Lower FILTER to around 3 kHz, add CHORUS, and a touch of DISTORTION for tape grit. On Flower EP, play harder for more “bark”: BRIGHT sets how much the velocity brings out.
3. **Psychedelic strings and choirs.** The Mellotron engine likes a slow ATTACK (0.1 to 0.4 s). WIDTH sets the depth of the tape wobble and TEXTURE its speed and hiss. Kaleidoscope sends a slow LFO to Timbre for an evolving texture.
4. **Afrobeats and Amapiano.** Daisy Kalimba, Marimba Blush and Echo Garden work well for log-drum-friendly melodies. Set a short ping-pong delay that follows the tempo: an eighth note is 30 / BPM seconds (0.27 s at 112 BPM).
5. **Trap and R&B pads.** Daisy Pad, Lavender Haze and Breeze Drift give soft beds for vocals. Set Filter Type to High Pass with FILTER around 200 to 300 Hz to keep the pad away from your 808. A slow Sine LFO on Filter (RATE 0.1 to 0.3 Hz) adds movement.
6. **Deep House organ.** Psyche Organ is ready for chord stabs. Shorten RELEASE for tight stabs. WIDTH sets the tremolo. Rotary Daisy adds a fast LFO on Timbre plus chorus for a swirling organ.
7. **Round bass lines.** The Bloom Bass presets play in Mono mode. Use GLIDE for slides (Slide Stem) and TEXTURE for drive (Fuzz Petal). Each note gets a slight random stereo offset: for a strictly centered bass, reduce the stereo separation of the mixer track.
8. **Expressive vibrato.** Set Destination to Pitch, AMOUNT to 0 and RATE between 5 and 6 Hz. Then ride the MODULATION wheel for vibrato on flute and violin presets. Automate the Mod Wheel parameter to repeat the gesture exactly.

MIDI, automation and performance

Notes and channels. CHINTZWAZ responds to all 128 MIDI notes on any MIDI channel. Velocity controls the level according to the VELOCITY slider. On the Flower EP engine, velocity also brightens the tone.

Polyphony. Up to 16 voices in Poly mode. When you play a 17th note, a voice is taken over and cut immediately. Long releases and delays on kalimba and pad presets use more voices. In Mono mode, each new note stops the previous one (last-note priority). Releasing the newest note does not bring back an earlier note that you still hold.

Pitch bend. MIDI pitch bend covers +/-2 semitones. The on-screen PITCH wheel adds another +/-2 semitones.

Mod wheel (CC1). CC1 raises the LFO depth, like the on-screen MODULATION wheel. When both are used, the higher value wins. CC1 acts on the notes that are sounding when the message arrives, so move the wheel while you play. For repeatable results in a song, automate the on-screen **Mod Wheel** parameter instead.

Other MIDI. The sustain pedal (CC64) is supported. All Notes Off and All Sound Off messages stop the sound. Aftertouch is ignored.

Automation in FL Studio. All 34 parameters can be automated. Move a control in the plugin, then use **Tools > Last tweaked > Create automation clip**. Use **Tools > Last tweaked > Link to controller** to assign a hardware knob. You can also pick parameters from the plugin wrapper menu (**Browse parameters**). Level, pitch, bend, filter, distortion and delay changes are smoothed to avoid clicks.

Latency and tail. The plugin adds no latency (0 samples), so no delay compensation is needed. It reports a 12-second tail to the host so reverb and delay trails are not cut when rendering.

CPU. CHINTZWAZ is lightweight. Effects set to zero are bypassed and cost no CPU. Engines with several oscillators per voice (Mellotron, Daisy Pad, Psyche Organ) cost a little more. Shorter releases reduce the number of sounding voices. The plugin works at 44.1, 48, 88.2 and 96 kHz.

FAQ

CHINTZWAZ does not appear in FL Studio. Open **Options > Manage plugins**, check that `C:\Program Files\Common Files\VST3` is in the search paths, then click **Find installed plugins**. The plugin is listed as **CHINTZWAZ** under **DJBILBOX BEATS**. It is an instrument: load it in the Channel rack, not in a Mixer insert slot.

I hear nothing, or a low rumble, when I click the on-screen keyboard. The on-screen keyboard covers the lowest MIDI notes (0 to 31). Most CHINTZWAZ sounds are designed for the middle and upper range. Play from a MIDI keyboard or the Piano roll, around C4 to C7 in FL Studio.

Where is the release of the filter envelope? There is no separate slider. The filter envelope release uses the same time as its DECAY slider in the ADSR+VELOCITY section. The filter envelope only acts when ENV is above zero.

The sound keeps moving even with AMOUNT at zero. Some engines have built-in movement: the Daisy Pad filter breathes slowly, and WIDTH sets the tape flutter (Mellotron), the vibrato (Hippie Flute) and the tremolo (Psyche Organ). Also check that the MODULATION wheel is at zero, because it adds LFO depth.

My chord only plays one note. The Mono/Poly menu is set to Mono, as on the flute leads and bass presets. Switch it to Poly.

CYBERPUNK LEAD

Neon lead synthesizer: supersaws, PWM, glitch FM, reese bass, laser plucks, brass, glass bells and bit-crushed FX.

Type	Synthesizer (VST3 instrument)
Format	VST3 64-bit (Windows)
Developer	DJBILBOX BEATS
Version	1.0
Factory presets	70
Latency	0 samples

Overview

CYBERPUNK LEAD is a neon, synthwave-flavored lead synthesizer by Humpire / DJBILBOX BEATS. It uses no samples. Eight engines cover the modern and retro-futuristic palette: a five-voice Neon Saw supersaw, a Chrome PWM with moving pulse width, a two-operator Glitch FM, a detuned Reese Bass, a Laser Pluck with a pitch dive, a Neon Brass section, a Data Crush engine with sample-rate and bit reduction, and a Hollow Glass FM bell.

Five character controls (TONE, BODY, BRIGHT, WIDTH, TEXTURE) reshape each engine. The sound then goes through a multimode filter with its own envelope, an amplitude envelope, an LFO and a master effects chain: distortion, chorus, ping-pong delay and reverb. The character is bright, wide and cutting, built to sit on top of a busy beat.

CYBERPUNK LEAD targets Trap and Drill leads, Synthwave and Retrowave, Hip-Hop hooks, Afrotrap melodies, G-Funk portamento leads, plus dark Reese bass lines and glitchy sound effects. It plays up to 16 notes at once. Most lead and bass presets load in Mono mode, many of them with a short glide.

Quick start in FL Studio

1. In the **Channel rack**, click **+** and choose **CYBERPUNK_LEAD** (vendor **DJBILBOX BEATS**). You can also drag it from the **Plugin database > Generators** browser.
2. The plugin opens on **CYBERPUNK LEAD - Init Patch**: the Neon Saw engine in Poly mode.
3. Click the **folder icon** at the left of the preset display. Pick a bank, then a preset. You can also step through all 70 presets with the **<** and **>** arrows.
4. Play from your MIDI keyboard or draw notes in the **Piano roll**. Leads sit well between C5 and C7 in FL Studio. The Reese & Bass presets are transposed one octave down: play them between C3 and C5.
5. Shape the sound with the three main controls:
   - **TONE**: the main character of the engine (sub level, PWM depth, FM ratio, pitch dive, crush rate...).
   - **FILTER**: the cutoff of the filter. Turn it down for a softer, darker lead.
   - **GLIDE**: slide time between notes. Leave it at zero for straight notes.
6. Adjust **VOLUME** to set the level, then click the **disk icon** to save your own version.

Interface

The window opens at 1000 × 746 pixels on a dark steel faceplate with pink and yellow neon accents. Drag the bottom-right corner to resize it from 50% to 200%. The proportions stay fixed.

Preset display (top center). From left to right: the folder icon (preset menu), the **<** arrow, the current preset name, the **>** arrow and the disk icon (save). The arrows wrap around from the last preset to the first.

EXPANSION cassette (top left). Clicking the cassette opens an information message about upcoming expansion packs. No expansion content is included with version 1.0.

Logo (top right). The DJBILBOXBEATS name and the DB logo.

Center. The camel artwork between two decorative speaker grilles, the “Cyberpunk Lead” title and “By Humpire”.

LFO section (left). RATE and **AMOUNT** set the LFO speed and depth. **PITCH**, also under this header, is the global transposition (-12 to +12 semitones). It is not an LFO control.

Amplitude envelope (left). Four sliders: **ATTACK, DECAY, SUSTAIN, RELEASE**.

Character knobs. A block of four small knobs next to the amplitude envelope: **TONE, BODY** (top) and **BRIGHT, WIDTH** (bottom). The fifth character control, **TEXTURE**, sits in a row with **RES** (filter resonance) and **ENV** (filter envelope depth) at the right of the FILTER knob. The **DELAY** knob (delay level) sits below RES.

GLIDE and FILTER. The two large knobs in the middle of the panel.

ADSR+VELOCITY section (upper right). Five sliders without individual labels. From left to right: filter envelope **Attack**, **Decay**, **Sustain**, **Release**, then **Velocity** (velocity sensitivity of the level). The value bubble shows the setting while you drag.

Output and effects (right). First row: **VOLUME, REVERB, CHORUS**. Second row: **DISTORTION, TIME** (delay time) and **FEEDBACK** (delay repeats).

Menu row. Five drop-down menus, from left to right: **Waveform** (engine), **LFO Source** (LFO shape), **Destination** (LFO target), **Filter Type** and **Mono/Poly**. Each menu shows its current value.

Wheels (bottom left). PITCH BEND springs back to the center when you release it. **MODULATION** stays where you leave it.

On-screen keyboard (bottom). Click a key to play it. Velocity follows the click position: click near the bottom of a key for a louder note. The keyboard starts at the lowest MIDI note (labeled C-2, which is C0 in FL Studio) and does not scroll. It covers MIDI notes 0 to 43, which is very low for lead sounds. Use a MIDI controller or the Piano roll for normal playing.

Mouse gestures. Drag a knob up or right to increase it. Drag sliders and wheels vertically. Double-click any knob or slider to return it to its default value. The mouse wheel also adjusts controls. A value bubble appears while you drag.

Parameters

Signal flow: engine, then filter, then amplitude envelope (per voice). The voices are summed, then processed by Distortion, Chorus, Delay and Reverb. A DC blocker and a soft limiter protect the output.

CYBERPUNK LEAD exposes 35 automatable parameters. The host name is shown in parentheses when it differs from the panel label.

Parameter	Range	Default	Function
Waveform (menu)	Neon Saw, Chrome PWM, Glitch FM, Reese Bass, Laser Pluck, Neon Brass, Data Crush, Hollow Glass	Neon Saw	Selects the synthesis engine.

Parameter	Range	Default	Function
TONE (Tone)	0 to 1	0.5	Character control 1. Its effect depends on the engine (see the table below).
BODY (Body)	0 to 1	0.5	Character control 2. Engine-dependent.
BRIGHT (Brightness)	0 to 1	0.5	Character control 3. Engine-dependent.
WIDTH (Width)	0 to 1	0.3	Character control 4. Engine-dependent (octave layer, PWM speed, feedback, detune...). It is not a stereo width control.
TEXTURE (Texture)	0 to 1	0.2	Character control 5. Engine-dependent (noise, sub level...).
FILTER (Filter)	40 to 18,000 Hz	9,000 Hz	Filter cutoff. The cutoff follows the keyboard by 25% around middle C.
RES (Resonance)	0 to 1	0.15	Filter resonance.
Filter Type (menu)	Low Pass, High Pass, Band Pass	Low Pass	Filter mode.
ENV (Filter Env)	0 to 1	0	Filter envelope depth. At 1 the envelope opens the cutoff up to 5 octaves. At 0 the filter envelope has no effect.
Slider 1, ADSR+VELOCITY (Filter Attack)	0.002 to 3 s	0.002 s	Filter envelope attack time.
Slider 2, ADSR+VELOCITY (Filter Decay)	0.02 to 4 s	0.3 s	Filter envelope decay time.
Slider 3, ADSR+VELOCITY (Filter Sustain)	0 to 1	0	Filter envelope sustain level.
Slider 4, ADSR+VELOCITY (Filter Release)	0.02 to 4 s	0.3 s	Filter envelope release time.
Slider 5, ADSR+VELOCITY (Velocity)	0 to 1	0.5	How much note velocity changes the level. 0: all notes at the same level. 1: full velocity response.
ATTACK (Attack)	0.001 to 5 s	0.01 s	Amplitude envelope attack time.
DECAY (Decay)	0.01 to 5 s	0.4 s	Amplitude envelope decay time.
SUSTAIN (Sustain)	0 to 1	0.7	Amplitude envelope sustain level.

Parameter	Range	Default	Function
RELEASE (Release)	0.02 to 8 s	0.4 s	Amplitude envelope release time.
VOLUME (Volume)	0 to 1	0.7	Level of each voice, before the effects.
GLIDE (Glide)	0 to 2	0	Portamento. The value is roughly the slide time in seconds. Each new note slides from the pitch of the previous note, in Mono and in Poly mode. 0 turns glide off.
PITCH (Pitch)	-12 to +12 semitones	0	Global transposition. The control is continuous: use whole values to stay in tune.
RATE (LFO Rate)	0.05 to 20 Hz	4 Hz	LFO speed. Not tempo-synced.
AMOUNT (LFO Amount)	0 to 1	0	LFO depth. The mod wheel adds to this value.
LFO Source (menu)	Sine, Triangle, Saw, Square, Sample&Hold	Sine	LFO shape. The LFO restarts at every new note. Sample&Hold picks a new random value once per cycle.
Destination (menu)	Pitch, Filter, Amp, Timbre	Pitch	LFO target. Pitch: up to +/-1 semitone (vibrato). Filter: up to +/-3 octaves (wobble). Amp: up to 50% level dip (tremolo). Timbre: moves all five character controls by up to +/-0.5.
Mono/Poly (menu)	Poly, Mono	Poly	Poly: up to 16 notes at once. Mono: one note at a time, last-note priority.
DISTORTION (Distortion)	0 to 1	0	Saturation on the summed output, with automatic level compensation.
CHORUS (Chorus)	0 to 1	0	Stereo chorus mix (slow, fixed rate).
DELAY (Delay)	0 to 1	0	Level of the ping-pong delay.
TIME (Delay Time)	0.05 to 1.0 s	0.35 s	Delay time in seconds. Not tempo-synced: for a quarter note, use 60 / BPM (0.5 s at 120 BPM); halve it for an eighth note.
FEEDBACK (Feedback)	0 to 0.9	0.35	Number of delay repeats (0 to 90% feedback).

Parameter	Range	Default	Function
REVERB (Reverb)	0 to 1	0.25	Reverb amount. Higher values also enlarge the room.
PITCH BEND wheel (Pitch Bend)	-1 to +1	0	On-screen pitch bend, +/-2 semitones. Adds to MIDI pitch bend.
MODULATION wheel (Mod Wheel)	0 to 1	0	On-screen mod wheel. Adds up to +0.8 to the LFO AMOUNT.

Character controls by engine

Engine	TONE	BODY	BRIGHT	WIDTH	TEXTURE
Neon Saw	Sub-octave level	Detune of the five saws	Built-in low-pass opening	Level of the octave-up saw	Noise
Chrome PWM	Pulse-width modulation depth	Detune of the second pulse	Built-in low-pass opening	Pulse-width modulation speed (0.3 to 5.3 Hz)	Sub-octave level
Glitch FM	Modulator ratio (0.5 to 4.0)	Decay time of the FM depth	FM depth	Modulator feedback	Attack noise burst
Reese Bass	Sub-octave level	Detune amount	Built-in low-pass opening	Saturation drive	Noise
Laser Pluck	Pitch dive at the attack (up to 2 octaves)	Length of the built-in filter sweep	Depth of the built-in filter sweep	Detune of the second saw	Attack noise
Neon Brass	Swell time of the built-in filter	Output weight	How far the filter opens	Detune of the three saws	Breath noise
Data Crush	Sample rate (higher is cleaner)	Bit depth (2 to 12 bits)	Built-in low-pass opening	Square-wave blend	Noise
Hollow Glass	Modulator ratio (1.4 to 4.5)	Decay length	FM depth (bell brightness)	Detune of the bell pair	Attack noise

The Hollow Glass engine has its own natural decay, like a struck bell. It fades out even while you hold the key. Raise BODY for a longer ring.

Factory presets

CYBERPUNK LEAD includes 70 factory presets in 7 banks of 10.

Neon Leads: Neon Rider, Chrome Hero, Night Drive, Hacker Lead, Laser Solo, Supersaw Wall, Soft Neon, Blade Lead, Octave Jump, Wire Lead.

Chrome & Pulse: Chrome Pulse, Wide PWM, Cyber Bass PWM, Fast Wobble, Retro Keys, Sunset Pad, Pulse Stab, Hollow Pulse, Arcade Lead, Pulse Drone.

Glitch FM: Glitch Bell, Data Pluck, FM Lead, Metal Keys, Digital Choir, Alien Voice, Cyber Marimba, Circuit Bass, Scan Lines, Neural Net.

Reese & Bass: Reese Core, Neon Sub, Dirty Reese, Wobble Grid, Industrial Bass, Cyber Slide, Acid Reese, Subway Bass, Bass Stab, Meltdown.

Laser Plucks: Laser Pluck, Neon Arp, Zap Down, Chrome Pluck, Techno Stab, Soft Pluck, Synth Harp, Cyber Koto, Sci-Fi Drop, Pluck Wall.

Neon Brass & Pads: Neon Brass, Blade Runner Pad, Synthwave Stab, Chrome Horns, Slow Open, Warm Neon, Hollow Glass, Ice Bell, Crystal Rain, Ghost Chime.

Data Crush & FX: Data Crush, 8-Bit Lead, Bit Rain, Glitch Bass, Corrupted Pad, Signal Lost, Modem Dial, Firewall, Static Wind, Boot Sequence.

Mono presets. These 32 presets load in Mono mode: all Neon Leads except Supersaw Wall; Cyber Bass PWM, Fast Wobble, Hollow Pulse and Arcade Lead; FM Lead, Alien Voice and Circuit Bass; all Reese & Bass presets; Data Crush, 8-Bit Lead, Glitch Bass, Signal Lost, Modem Dial and Firewall. All other presets load in Poly mode.

Loading a preset first resets every parameter to its default value, then applies the preset. The two wheels are not changed.

User presets. Click the disk icon to save the current sound as a `.cppreset` file. The default folder is `Documents\DJBILBOX BEATS\CYBERPUNK_LEAD\Presets`. Files in this folder appear in the **User** submenu of the preset menu. Use **Open a .cppreset file...** in the same menu to load a file from another location. The preset display shows the file name.

The host also sees the 70 factory presets as a program list. Your project saves the full state of the plugin, including the preset name.

Blade Runner is a trademark of its owner. It is used here only to describe a sound style. DJBILBOX BEATS is not affiliated with the trademark owner.

Production tips

1. **Trap and Drill leads.** Start from Neon Rider or Night Drive (Mono, short glide). A dotted-eighth delay is 45 / BPM seconds (0.375 s at 120 BPM, as in Neon Rider). Lower DELAY when the vocal comes in.
2. **G-Funk portamento lead.** Load Soft Neon or Hollow Pulse in Mono mode. Lower BODY (less detune) and BRIGHT, set FILTER between 2 and 4 kHz and GLIDE between 0.1 and 0.2. Add a gentle vibrato: Destination Pitch, Sine LFO, RATE 5 to 6 Hz, AMOUNT around 0.06 (as in Chrome Hero). Play high, around C6 to C7 in FL Studio.
3. **Synthwave chords and pads.** Supersaw Wall, Sunset Pad, Warm Neon and Synthwave Stab play chords in Poly mode. On Neon Saw, BODY widens the supersaw detune and WIDTH adds an octave-up layer. Add CHORUS and REVERB for the classic 80s wash.
4. **Plucks and arps for Afrotrap and Trap.** Laser Pluck, Neon Arp and Chrome Pluck react well to short notes. TONE sets the “laser” pitch dive. Combine them with FL Studio’s channel arpeggiator (Channel settings, Miscellaneous functions tab) and a tempo-matched delay.
5. **Dark Reese bass.** The Reese & Bass presets are in Mono mode and transposed one octave down. BODY sets the detune and WIDTH drives the saturation. Wobble Grid sends a Sine LFO to the filter. Each note gets a slight random stereo offset: for a strictly centered bass, reduce the stereo separation of the mixer track.
6. **FM bells and glitch keys.** On Glitch FM and Hollow Glass, TONE changes the frequency ratio. Some positions sound harmonic, others metallic and inharmonic: sweep TONE slowly and stop where the bell fits your key. BODY sets how long the bright attack lasts.
7. **8-bit and lo-fi FX.** On Data Crush, lower TONE for a lower sample rate and lower BODY for fewer bits (8-Bit Lead). Use FILTER to tame harsh aliasing. The crushing is lighter at 88.2 and 96 kHz project rates: lower TONE further for the same effect.

8. **Expressive vibrato.** Set Destination to Pitch, AMOUNT to 0 and RATE between 5 and 6 Hz. Then ride the MODULATION wheel at the end of long notes. Automate the Mod Wheel parameter to repeat the gesture exactly.

MIDI, automation and performance

Notes and channels. CYBERPUNK LEAD responds to all 128 MIDI notes on any MIDI channel. Velocity controls the level according to the Velocity slider (fifth slider of the ADSR+VELOCITY section).

Polyphony. Up to 16 voices in Poly mode. When you play a 17th note, a voice is taken over and cut immediately. In Mono mode, each new note stops the previous one (last-note priority). Releasing the newest note does not bring back an earlier note that you still hold.

Pitch bend. MIDI pitch bend covers +/-2 semitones. The on-screen PITCH BEND wheel adds another +/-2 semitones.

Mod wheel (CC1). CC1 raises the LFO depth, like the on-screen MODULATION wheel. When both are used, the higher value wins. CC1 acts on the notes that are sounding when the message arrives, so move the wheel while you play. For repeatable results in a song, automate the on-screen **Mod Wheel** parameter instead.

Other MIDI. The sustain pedal (CC64) is supported. All Notes Off and All Sound Off messages stop the sound. Aftertouch is ignored.

Automation in FL Studio. All 35 parameters can be automated. Move a control in the plugin, then use **Tools > Last tweaked > Create automation clip**. Use **Tools > Last tweaked > Link to controller** to assign a hardware knob. You can also pick parameters from the plugin wrapper menu (**Browse parameters**). Level, pitch, bend, filter, distortion and delay changes are smoothed to avoid clicks.

Latency and tail. The plugin adds no latency (0 samples), so no delay compensation is needed. It reports a 12-second tail to the host so reverb and delay trails are not cut when rendering.

CPU. CYBERPUNK LEAD is lightweight. Effects set to zero are bypassed and cost no CPU. The Neon Saw engine (seven oscillators per voice) costs the most in Poly mode; Mono mode and shorter releases keep the voice count low. The plugin works at 44.1, 48, 88.2 and 96 kHz.

FAQ

CYBERPUNK LEAD does not appear in FL Studio. Open **Options > Manage plugins**, check that `C:\Program Files\Common Files\VST3` is in the search paths, then click **Find installed plugins**. The plugin is listed as **CYBERPUNK_LEAD** under **DJBILBOX BEATS**. It is an instrument: load it in the Channel rack, not in a Mixer insert slot.

I hear nothing, or a low rumble, when I click the on-screen keyboard. The on-screen keyboard covers the lowest MIDI notes (0 to 43). Lead sounds are designed for the middle and upper range. Play from a MIDI keyboard or the Piano roll, around C5 to C7 in FL Studio.

Which slider is which in the ADSR+VELOCITY section? From left to right: filter envelope Attack, Decay, Sustain, Release, then Velocity. The first four only change the sound when ENV is above zero.

My bell preset fades out while I hold the key. The Hollow Glass engine (Hollow Glass, Ice Bell, Crystal Rain, Ghost Chime) has a built-in bell decay. Raise BODY for a longer ring, or switch to Neon Brass or Chrome PWM for a sustained sound.

My chord only plays one note. The Mono/Poly menu is set to Mono, as on most lead and bass presets. Switch it to Poly.

DZPOWER

Algerian Raï, Chaabi and Casbah sounds from a pure synthesis engine: plucked strings, reed flutes, organ, strings and hand drums.

Type	Synthesizer (VST3 instrument)
Format	VST3 64-bit (Windows)
Developer	DJBILBOX BEATS
Version	1.0
Factory presets	70
Latency	0 samples

Overview

DZPOWER is a 16-voice synthesizer dedicated to the sounds of Algerian popular music. It has eight synthesis engines inspired by traditional instruments: Mandole (Algerian mandolin), Qanun (zither), Gasba (reed flute), Ghaita (double-reed zorna), Rai Organ, Chaabi Strings, Derbouka (goblet drum) and Bendir Sub (frame drum turned into a deep sub). It uses no samples. Every sound is generated in real time, so it reacts to the five timbre knobs, the filter and the envelopes.

Each engine runs through a resonant multimode filter with its own envelope, a key-synced LFO and an effects chain: distortion, chorus, stereo ping-pong delay and reverb. A soft limiter at the output keeps the level under control.

DZPOWER is built for Raï and Raï'n'B, Chaabi, Oriental pop, Afrotrap, Hip-Hop and Trap. Use the Mandole and Qanun for melodic hooks, the Gasba and Ghaita for lead lines, the Rai Organ for classic keyboard parts, the strings for pads, the Derbouka for percussion and the Bendir Sub as an 808-style bass.

Quick start in FL Studio

1. In the **Channel rack**, click **+** and choose **DZPOWER** (manufacturer **DJBILBOX BEATS**). You can also drag it from **Plugin database > Generators**.
2. The interface opens on **DZPOWER - Init Patch**.
3. Click the folder icon at the left of the preset display. Open a bank (for example **Mandole**) and pick a preset. You can also step through all 70 presets with the **<** and **>** arrows.
4. Play notes from your MIDI keyboard, or draw them in the **Piano roll**.
5. Turn **CUTOFF** to make the sound darker or brighter.
6. Turn **BRIGHT** (in the left speaker grille) to change the brightness of the engine itself.
7. Turn **REVERB** to place the sound in a small or large space.
8. Click the disk icon at the right of the preset display to save your own version.

Interface

The window opens at 1000 × 746 pixels. Drag the bottom-right corner to resize it from half size to double size. The proportions stay fixed.

Preset display (top center). From left to right: the folder icon opens the preset menu, **<** loads the previous preset, the center shows the current preset name, **>** loads the next preset, and the disk icon saves a user preset. The preset menu lists the seven factory banks as submenus, a **User** submenu with your saved presets, and an **Open a**

.dzpreset file... item.

EXPANSION cassette (top left). Click it to see information about upcoming expansion packs (**EXPANSION VOL.1**) and the link to the DJBILBOX BEATS store.

Logo (top right). The DJBILBOX BEATS logo and name.

Speaker grilles and emblem (center). The camel emblem sits between two perforated speaker grilles that hold the small knobs. The left grille holds **TONE**, **BODY**, **BRIGHT** and **WIDTH**. These four knobs and **TEXTURE** shape the selected engine. The right grille holds **TEXTURE**, **ENV** (filter envelope depth) and **DELAY** (delay level).

VOLUME and REVERB (top right). Level of the instrument and amount of reverb.

ASDR group (right). Five vertical sliders under the "ASDR" heading: **ATTACK**, **DECAY**, **SUSTAIN** and **RELEASE** of the filter envelope, plus **VELOCITY** (velocity sensitivity of the volume).

LFO group (left). RATE and **AMOUNT** set the LFO speed and depth. **PITCH** sits in the same row but is the global transpose of the instrument.

Amplitude envelope (left, below the LFO). Four vertical sliders: **ATTACK**, **DECAY**, **SUSTAIN** and **RELEASE** of the volume envelope.

Center row. GLIDE, then the **FILTER** group with **CUTOFF** and **RESONANCE**, then **CHORUS**, **DISTORTION**, and the **DELAY** group with **TIME** and **FEEDBACK**.

Menu row. Five drop-down menus. Each one shows its current value. From left to right: engine (Waveform), LFO shape (LFO Source), LFO target (Destination), filter mode (Filter Type) and voice mode (Poly / Mono).

Wheels (bottom left). Pitch Bend springs back to center when you release the mouse. **Modulation** stays where you leave it.

On-screen keyboard (bottom). Click a key to audition the sound. It also lights up incoming MIDI notes within its range. In version 1.0 the keyboard covers the lowest part of the MIDI range (keys labeled C-2 to about A0, MIDI notes 0 to 33). Use it to check bass and sub notes. Play melodies and chords from a MIDI keyboard or the Piano roll.

Using the controls. Drag a knob up/down or left/right. Drag a slider vertically. The mouse wheel adjusts in small steps. Double-click to return to the default value. Hold Ctrl or Alt while you drag for velocity-sensitive, finer movement. A small bubble shows the current value while you hover or drag.

Parameters

Names in parentheses are the parameter names shown in FL Studio for automation, when they differ from the panel label. "Amp" means the amplitude (volume) envelope.

Parameter	Range	Default	Function
Engine menu (Waveform)	Mandole, Qanun, Gasba, Ghaita, Rai Organ, Chaabi Strings, Derbouka, Bendir Sub	Mandole	Selects the synthesis engine. See the engine table below.
LFO shape menu (LFO Source)	Sine, Triangle, Saw, Square, Sample&Hold	Sine	LFO waveform. Sample&Hold picks a new random value once per cycle.

Parameter	Range	Default	Function
LFO target menu (Destination)	Pitch, Filter, Amp, Timbre	Pitch	Where the LFO goes. Pitch: up to ±1 semitone. Filter: up to ±3 octaves of cutoff. Amp: tremolo, up to 50% volume dip. Timbre: moves all five timbre knobs together, up to ±0.5.
Filter mode menu (Filter Type)	Low Pass, High Pass, Band Pass	Low Pass	12 dB/octave state-variable filter mode. Band Pass gets a small level boost.
Voice mode menu (Mono/Poly)	Poly, Mono	Poly	Poly: up to 16 notes. Mono: each new note releases the previous one.
RATE (LFO Rate)	0.05 to 20 Hz	4 Hz	LFO speed. The LFO restarts at each new note. It is not tempo-synced.
AMOUNT (LFO Amount)	0 to 1	0	LFO depth. The Modulation wheel (or MIDI CC1) adds up to 0.8 on top.
PITCH (Pitch)	-12 to +12 semitones, continuous	0	Global transpose. Values between semitones detune the whole instrument.
GLIDE (Glide)	0 to 2 s	0 (off)	Portamento. Above 0, each new note slides from the pitch of the previous note, in Poly and Mono.
CUTOFF (Filter)	40 Hz to 18 kHz	9 kHz	Filter cutoff frequency. The cutoff follows the played note at 25% (key tracking).
RESONANCE (Resonance)	0 to 1	0.15	Filter resonance (emphasis at the cutoff).
ENV (Filter Env)	0 to 1	0	Filter envelope depth. At 1, the envelope opens the cutoff by up to 5 octaves.
ASDR ATTACK (Filter Attack)	0.002 to 3 s	0.002 s	Filter envelope attack time.
ASDR DECAY (Filter Decay)	0.02 to 4 s	0.3 s	Filter envelope decay time.
ASDR SUSTAIN (Filter Sustain)	0 to 1	0	Filter envelope sustain level.
ASDR RELEASE (Filter Release)	0.02 to 4 s	0.3 s	Filter envelope release time.
VELOCITY (Velocity)	0 to 1	0.5	How much note velocity changes the volume. 0 = every note at full level. 1 = full velocity response.
Amp ATTACK (Attack)	0.001 to 5 s	0.01 s	Volume envelope attack time.

Parameter	Range	Default	Function
Amp DECAY (Decay)	0.01 to 5 s	0.4 s	Volume envelope decay time.
Amp SUSTAIN (Sustain)	0 to 1	0.7	Volume envelope sustain level. Set to 0 for plucks and drums.
Amp RELEASE (Release)	0.02 to 8 s	0.4 s	Volume envelope release time after note-off.
TONE (Tone)	0 to 1	0.5	Engine timbre control 1. See the engine table.
BODY (Body)	0 to 1	0.5	Engine timbre control 2. See the engine table.
BRIGHT (Brightness)	0 to 1	0.5	Engine timbre control 3. See the engine table.
WIDTH (Width)	0 to 1	0.3	Engine timbre control 4. See the engine table.
TEXTURE (Texture)	0 to 1	0.2	Engine timbre control 5. See the engine table.
VOLUME (Volume)	0 to 1	0.7	Instrument level, before the effects.
DISTORTION (Distortion)	0 to 1	0	Soft saturation. Drive rises up to 15x; the level is compensated.
CHORUS (Chorus)	0 to 1	0	Chorus mix (fixed 0.7 Hz modulation).
DELAY (Delay)	0 to 1	0	Level of the stereo ping-pong delay.
TIME (Delay Time)	0.05 to 1 s	0.35 s	Delay time in seconds.
FEEDBACK (Feedback)	0 to 0.9	0.35	Delay repeats. Higher values give more echoes.
REVERB (Reverb)	0 to 1	0.25	Reverb amount. The room also grows larger as you turn it up.
Pitch Bend wheel (Pitch Bend)	-1 to +1	0	Bends all notes by up to ±2 semitones. Adds to the MIDI pitch wheel.
Modulation wheel (Mod Wheel)	0 to 1	0	Adds LFO depth (up to +0.8). The higher of this wheel and MIDI CC1 is used.

Effects order: Distortion, Chorus, Delay, Reverb, then the output soft limiter. Each effect is bypassed while its knob is at zero.

What the timbre knobs do on each engine

Engine	TONE	BODY	BRIGHT	WIDTH	TEXTURE
Mandole: plucked string with wooden body	Pick hardness and body resonance pitch	String sustain	String brightness (damping)	No effect	Wooden body resonance
Qanun: bright plucked zither with a sympathetic octave	No effect	String sustain	String brightness (damping)	Level of the sympathetic octave at the attack	No effect
Gasba: reed flute with breath and vibrato	Color of the breath noise	Focus of the breath noise	2nd and 3rd harmonics	Vibrato depth (fades in during the first half second)	Breath level
Ghaita: double reed with nasal resonances	Position of the nasal resonances	Strength of the nasal resonances	Brightness	Detune between the two reeds	Breath noise
Rai Organ: drawbars, tremolo, key click	Sub-octave drawbar	Octave drawbar	Upper harmonic drawbars	Tremolo depth and speed	Key click
Chaabi Strings: five detuned saws with delayed vibrato	Body resonance pitch	Body resonance level	Brightness	Vibrato depth (starts after 0.2 s)	Bow noise
Derbouka: doum tone and tak slap	Pitch drop of the doum	Doum decay	Pitch of the slap	Level of the tak slap	Attack click
Bendir Sub: deep sub with jingles	Knock (pitch punch at the attack)	2nd harmonic	Rattle decay	Saturation	Rattle level

Factory presets

DZPOWER includes 70 factory presets in 7 banks of 10. Loading a factory preset resets every parameter before applying the preset. The Pitch Bend and Modulation wheels are not changed.

Mandole Algiers Mandole, Mandole Chaabi, Casbah Pick, Warm Mandole, Tremolo Pick, Low Mandole, Bright Strum, Rai Pluck, Muted Mandole, Desert Night Pluck.

Qanun & Plucks Qanun Dar, Zither Rain, Harp Of Tlemcen, Bright Qanun, Soft Qanun, Oriental Arp, Low Qanun, Glass Qanun, Sahara Strings, Zellij Pluck.

Gasba & Ghaita Gasba Dream, Desert Gasba, Breathy Gasba, Sufi Gasba, High Gasba, Ghaita Call, Zorna Celebration, Nasal Reed, Mizmar Dz, Snake Zorna.

Rai Organ Rai Organ, Oran Night, Full Rai, Rai Lead Organ, Accordion Dz, Vintage Rai Keys, Soft Flute Stop, Rotary Casbah, Tape Rai, Golden Swell.

Chaabi Strings Chaabi Strings, Algiers Violin, Slow Swell, Dark Cellos, High Violins, Pizzicato Dz, Andalusian Pad, Wobble Violin, Suspense Casbah, Rooftop Strings.

Derbouka & Perc Derbouka Doum, Tak Tak, Deep Doum, Slap Derbouka, Tabla Dz, Rai Percussion, Dub Doum, Hand Drum Fill, Crowd Clap, Drop Doum.

Bendir Sub & Bass Bendir Sub, Deep Casbah, Rattle Sub, Long Slide Sub, Pitch Drop Dz, Dirty Bendir, Smooth Sub, Hard Knock Dz, Short Boom, Algiers Rumble.

User presets. The disk icon saves the current sound as a `.dzpreset` file in `Documents\DJBILBOX BEATS\DZPOWER\Presets`. Files in this folder appear in the **User** submenu of the preset menu. Use **Open a .dzpreset file...** to load a preset from another folder. Your sound is also saved inside your FL Studio project.

Production tips

- **Raï lead vibrato.** On most presets the LFO is set to Sine, Pitch, 5 Hz with AMOUNT at 0. Push the Modulation wheel to add vibrato only where you want it. Try **Rai Lead Organ** or **Zorna Celebration**: both are in Mono with a short glide for legato phrases.
- **Quarter-tone inflections.** DZPOWER plays equal temperament. The pitch bend range is ±2 semitones, so a quarter tone is one quarter of the upward wheel travel. For a whole passage, set **PITCH** to +0.5 or -0.5. The knob is continuous.
- **Delay in time with the beat.** TIME is set in seconds. Quarter note = 60 / BPM. Eighth note = 30 / BPM. Dotted eighth = 45 / BPM. At 100 BPM, an eighth note is 0.30 s. At 140 BPM, it is 0.21 s.
- **Afrotrap mandole hooks.** Start from **Casbah Pick** or **Algiers Mandole**. Raise **VELOCITY** to 0.8 or more so that your playing dynamics come through. Lower **BODY** for shorter, tighter notes that leave room for the drums.
- **808-style slides with the Bendir Sub.** Use **Long Slide Sub** or set GLIDE between 0.1 and 0.35 in Mono. Any GLIDE above 0 makes every note slide from the previous one, so set it back to 0 for straight notes. Raise **BODY** to add a 2nd harmonic that stays audible on phone speakers. Add **DISTORTION** around 0.2 for grit.
- **Derbouka grooves.** Each hit combines a pitched doum and a slap. Play low notes for deep doums and higher notes for tighter hits; use **Tak Tak** or **Slap Derbouka** for slap accents. **TONE** controls the pitch drop of the doum. **WIDTH** controls the slap level. Keep Poly mode so that hits overlap naturally. Vary note velocities in the Piano roll for a human feel.
- **Chaabi string pads.** The vibrato of the Chaabi Strings engine starts 0.2 s after the note begins. Short notes stay straight, long notes bloom. Add CHORUS around 0.3 and REVERB around 0.5 for width. **Suspense Casbah** uses a slow filter envelope for intros and builds.
- **Keep chords clean.** The output passes through a soft limiter. Dense chords at high VOLUME saturate gently. Lower **VOLUME** for big chords, and raise it on plucks and drums, which peak higher than they sound.

MIDI, automation and performance

- **Notes.** DZPOWER responds to all MIDI notes (0 to 127) on all MIDI channels, tuned to equal temperament with A4 = 440 Hz. Middle C is MIDI note 60 (C5 in FL Studio).
- **Polyphony.** 16 voices in Poly mode. When you play more than 16 notes, the oldest note is stolen; the lowest and highest held notes are protected. In Mono mode, each new note releases the previous one (its release tail still sounds). Releasing a note does not bring back an earlier held note.
- **Velocity.** Velocity changes the volume according to the VELOCITY slider.
- **Pitch bend.** ±2 semitones. The MIDI pitch wheel and the on-screen wheel add together.
- **Modulation (CC1).** Adds up to 0.8 to the LFO AMOUNT. The higher of MIDI CC1 and the on-screen wheel is used.
- **Sustain pedal (CC64).** Supported. Aftertouch and other MIDI CCs are ignored. Use host automation for everything else.
- **Automation in FL Studio.** Every parameter in the table can be automated. Move a control on the DZPOWER panel, then choose **Tools > Last tweaked > Create automation clip**. To use a hardware knob, choose **Tools > Last tweaked > Link to controller** and move the knob. Volume, pitch, bend and cutoff changes are smoothed to avoid zipper noise.
- **Latency.** 0 samples. No delay compensation is needed. The plugin reports a 12-second tail to the host so that delay and reverb tails are not cut when you render.

- **CPU.** DZPOWER uses no samples and streams nothing from disk. The CPU load grows with the number of sounding notes. The five-saw Chaabi Strings engine is the heaviest. Effects at zero are bypassed and use no CPU. Long release times keep voices active longer.
- **Sample rates.** 44.1, 48, 88.2 and 96 kHz.

FAQ

I hear no sound. Check that the DZPOWER channel is routed to a Mixer track and that the channel is not muted. Check VOLUME. With Low Pass selected, a very low CUTOFF can silence the sound. With High Pass selected, a high CUTOFF removes bass presets. Finally, remember that the on-screen keyboard plays very low notes in version 1.0: test with a MIDI keyboard or the Piano roll.

DZPOWER does not appear in FL Studio. DZPOWER is a 64-bit VST3 plugin. Check that `DZPOWER.vst3` is in `C:\Program Files\Common Files\VST3`. In FL Studio, open **Options > Manage plugins** and click **Find installed plugins**. Look for it under **Generators**, manufacturer **DJBILBOX BEATS**.

Plucks and drums sound quieter than pads. Plucked and percussive presets have sharp attacks and short notes, so they sound softer at the same peak level. Raise **VOLUME** on these presets.

Can DZPOWER play maqam scales with quarter tones? There is no built-in scale or microtuning. Use the pitch wheel (a quarter tone is a quarter of the upward travel) or the continuous **PITCH** knob for inflections.

The sound gets quieter when I turn REVERB fully down. In version 1.0, the reverb stage also lifts the dry signal by about 6 dB while it is active. At 0 the reverb is bypassed, so the overall level drops. For a dry sound at the usual level, leave REVERB at a very low value such as 0.01, or compensate with VOLUME.

FUNKYKOPS

Funk, disco and G-Funk keys, basses, brass, strings and talkbox leads from a pure synthesis engine.

Type	Synthesizer (VST3 instrument)
Format	VST3 64-bit (Windows)
Developer	DJBILBOX BEATS
Version	1.0
Factory presets	70
Latency	0 samples

Overview

FUNKYKOPS is a 16-voice synthesizer for the classic sounds of funk, disco and West Coast G-Funk. It has eight synthesis engines: Clavinet (percussive plucked keys), Rhodes (electric piano), Synth Bass, Brass, Strings, Lead, Guitar and Talkbox. It uses no samples. Every note is generated in real time and responds to the timbre knobs, the filter and its envelope.

Each engine runs through a resonant multimode filter with a fast decay envelope, the classic tool for funky bass “quacks”, clav wah and brass stabs. A key-synced LFO, glide and a Mono mode cover whining G-Funk leads. The effects chain adds distortion, chorus, a stereo ping-pong delay and reverb. A soft limiter at the output keeps the level under control.

FUNKYKOPS is built for G-Funk, Funk, Disco and Boogie, Hip-Hop, Nu-Disco and Deep House. Use the Clavinet and Guitar for rhythm parts, the Rhodes for chords, the Synth Bass for bass lines, the Brass and Strings for stabs and pads, and the Lead and Talkbox engines for melodies.

Quick start in FL Studio

1. In the **Channel rack**, click **+** and choose **FUNKYKOPS** (manufacturer **DJBILBOX BEATS**). You can also drag it from **Plugin database > Generators**.
2. The interface opens on **FUNKYKOPS - Init Patch**.
3. Click **Presets** at the left of the preset display. Open a bank (for example **Synth Bass**) and pick a preset. You can also step through all 70 presets with the **<** and **>** buttons.
4. Play notes from your MIDI keyboard, or draw them in the **Piano roll**.
5. Turn **FILTER** to make the sound darker or brighter.
6. Turn **BRIGHT** to change the brightness of the engine itself.
7. Turn **CHORUS** for width and a vintage shimmer.
8. Click **Save** to store your own version.

Interface

The window opens at 1000 × 746 pixels. Drag the bottom-right corner to resize it from half size to double size. The proportions stay fixed.

Preset display (top center). From left to right: the **Presets** button opens the preset menu, the center shows the current preset name, **<** and **>** load the previous and next presets, and **Save** saves a user preset. The preset menu lists the seven factory banks as submenus, a **User** submenu with your saved presets, and an **Open a .fkpreset file...** item.

Cassette (top left). Click it to see information about upcoming expansion packs (**EXPANSION VOL.1**) and the link to the DJBILBOX BEATS store.

Logo (top right). The DJBILBOX BEATS logo.

Emblem and speaker grilles (center). The camel emblem sits between two decorative speaker grilles. The FUNKYKOPS title and “By Humpire” are below the emblem.

LFO group (left). RATE and **AMOUNT** set the LFO speed and depth. **PITCH** sits in the same row but is the global transpose of the instrument.

Amplitude envelope (left, below the LFO). Four vertical sliders: **ATTACK**, **DECAY**, **SUSTAIN** and **RELEASE** of the volume envelope.

FILTER ENV + VELOCITY group (top right). Four vertical sliders: **ENV** (filter envelope depth), **DEC** (filter envelope decay), **RES** (filter resonance) and **VEL** (velocity sensitivity of the volume). **VOLUME** is to their right.

Center row. From left to right: the small **VOWEL** and **BODY** knobs, **GLIDE**, **FILTER** (cutoff), then the small **BRIGHT**, **WIDTH** and **TEXTURE** knobs. The five small knobs shape the selected engine.

Effects (right). REVERB, **CHORUS** and **DISTORTION** on the first row. **DELAY** (level), **TIME** and **FEEDBACK** on the second row.

Menu row. Five drop-down menus. Each one shows its current value. From left to right: engine (Waveform), LFO shape (LFO Source), LFO target (Destination), filter mode (Filter Type) and voice mode (Poly / Mono).

Wheels (bottom left, “Pitch Bend / Mod”). The left wheel bends the pitch and springs back to center when you release the mouse. The right wheel is the modulation wheel and stays where you leave it.

On-screen keyboard (bottom). Click a key to audition the sound. It also lights up incoming MIDI notes within its range. In version 1.0 the keyboard covers the lowest part of the MIDI range (keys labeled C-2 to about D1, MIDI notes 0 to 38). Use it to check bass notes. Play melodies and chords from a MIDI keyboard or the Piano roll.

Using the controls. Drag a knob up/down or left/right. Drag a slider vertically. The mouse wheel adjusts in small steps. Double-click to return to the default value. Hold Ctrl or Alt while you drag for velocity-sensitive, finer movement. A small bubble shows the current value while you hover or drag.

Parameters

Names in parentheses are the parameter names shown in FL Studio for automation, when they differ from the panel label.

Parameter	Range	Default	Function
Engine menu (Waveform)	Clavinet, Rhodes, Synth Bass, Brass, Strings, Lead, Guitar, Talkbox	Clavinet	Selects the synthesis engine. See the engine table below.
LFO shape menu (LFO Source)	Sine, Triangle, Saw, Square, Sample&Hold	Sine	LFO waveform. Sample&Hold picks a new random value once per cycle.

Parameter	Range	Default	Function
LFO target menu (Destination)	Pitch, Filter, Amp, Timbre	Pitch	Where the LFO goes. Pitch: up to ±1 semitone. Filter: up to ±3 octaves of cutoff (auto-wah). Amp: tremolo, up to 50% volume dip. Timbre: moves all five timbre knobs together, up to ±0.5.
Filter mode menu (Filter Type)	Low Pass, High Pass, Band Pass	Low Pass	12 dB/octave state-variable filter mode. Band Pass gets a small level boost.
Voice mode menu (Mono/Poly)	Poly, Mono	Poly	Poly: up to 16 notes. Mono: each new note releases the previous one.
RATE (LFO Rate)	0.05 to 20 Hz	4 Hz	LFO speed. The LFO restarts at each new note. It is not tempo-synced.
AMOUNT (LFO Amount)	0 to 1	0	LFO depth. The modulation wheel (or MIDI CC1) adds up to 0.8 on top.
PITCH (Pitch)	-12 to +12 semitones, continuous	0	Global transpose. Values between semitones detune the whole instrument.
GLIDE (Glide)	0 to 2 s	0 (off)	Portamento. Above 0, each new note slides from the pitch of the previous note, in Poly and Mono.
FILTER (Filter)	40 Hz to 18 kHz	9 kHz	Filter cutoff frequency. The cutoff follows the played note at 25% (key tracking).
RES (Resonance)	0 to 1	0.15	Filter resonance (emphasis at the cutoff).
ENV (Filter Env)	0 to 1	0	Filter envelope depth. At 1, the envelope opens the cutoff by up to 5 octaves.
DEC (Filter Decay)	0.02 to 2 s	0.3 s	Filter envelope decay. The envelope has a fixed 2 ms attack and no sustain; it falls back over the DEC time, also after note-off.
VEL (Velocity)	0 to 1	0.5	How much note velocity changes the volume. 0 = every note at full level. 1 = full velocity response.
ATTACK (Attack)	0.001 to 5 s	0.01 s	Volume envelope attack time.
DECAY (Decay)	0.01 to 5 s	0.4 s	Volume envelope decay time.
SUSTAIN (Sustain)	0 to 1	0.7	Volume envelope sustain level.

Parameter	Range	Default	Function
RELEASE (Release)	0.02 to 8 s	0.4 s	Volume envelope release time after note-off.
VOWEL (Vowel / Attack)	0 to 1	0.5	Engine timbre control 1. Active on the Talkbox engine only. See the engine table.
BODY (Body / Resonance)	0 to 1	0.5	Engine timbre control 2. See the engine table.
BRIGHT (Brightness)	0 to 1	0.5	Engine timbre control 3. See the engine table.
WIDTH (Width / Detune)	0 to 1	0.3	Engine timbre control 4. See the engine table.
TEXTURE (Texture)	0 to 1	0.2	Engine timbre control 5. See the engine table.
VOLUME (Volume)	0 to 1	0.7	Instrument level, before the effects.
DISTORTION (Distortion)	0 to 1	0	Soft saturation. Drive rises up to 15x; the level is compensated.
CHORUS (Chorus)	0 to 1	0	Chorus mix (fixed 0.7 Hz modulation).
DELAY (Delay)	0 to 1	0	Level of the stereo ping-pong delay.
TIME (Delay Time)	0.05 to 1 s	0.35 s	Delay time in seconds.
FEEDBACK (Feedback)	0 to 0.9	0.35	Delay repeats. Higher values give more echoes.
REVERB (Reverb)	0 to 1	0.25	Reverb amount. The room also grows larger as you turn it up.
Pitch bend wheel (Pitch Bend)	-1 to +1	0	Bends all notes by up to ±2 semitones. Adds to the MIDI pitch wheel.
Modulation wheel (Mod Wheel)	0 to 1	0	Adds LFO depth (up to +0.8). The higher of this wheel and MIDI CC1 is used.

Effects order: Distortion, Chorus, Delay, Reverb, then the output soft limiter. Each effect is bypassed while its knob is at zero.

What the timbre knobs do on each engine

Engine	VOWEL	BODY	BRIGHT	WIDTH	TEXTURE
Clavinet: plucked string with pickup bite	No effect	String sustain	String brightness (damping)	No effect	Pickup bite (saturation)

Engine	VOWEL	BODY	BRIGHT	WIDTH	TEXTURE
Rhodes: tine and bell partials, natural decay	No effect	Decay length	Bell “bark” amount (grows with velocity)	No effect	Hammer attack noise
Synth Bass: saw/square with a sub-octave sine	No effect	No effect	Oscillator brightness	Sub-octave level	Saw-to-square blend
Brass: three detuned saws	No effect	No effect	Brightness	Detune	Breath noise
Strings: five-saw ensemble	No effect	No effect	Brightness	Detune (ensemble width)	No effect
Lead: two saws plus a square one octave below	No effect	No effect	Brightness	Detune	No effect
Guitar: short plucked string	No effect	String sustain	String brightness (damping)	No effect	No effect
Talkbox: three saws through two vowel formants	Vowel (from a dark “oo” to an open vowel)	Formant strength	Brightness	Detune	Breath noise

Factory presets

FUNKYKOPS includes 70 factory presets in 7 banks of 10. Loading a factory preset resets every parameter before applying the preset. The pitch bend and modulation wheels are not changed.

Clavinet & Funk Keys Chicken Scratch, Disco Clav, Wah Patrol, Afro Clav, Stevie Stab, Funk Pickup, Bootsy Clav, Siren Clav, Dusty Clav, Badge Clav.

Rhodes & EP Rhodes Suitcase, Silk Rhodes, Bark Rhodes, Midnight EP, Disco EP, Tremolo EP, Slow Jam Keys, Wurly Bite, Bell EP, Lowrider Keys.

Synth Bass Funk Moog, Dre Bass, Rubber Bass, Sub Cop, Slap Synth, G-Funk Whine Bass, Disco Octave, Acid Funk, Bus Driver, Mean Streets.

Brass & Stabs Horn Section, Funk Stab, Police Horns, Disco Brass, Swell Brass, Soft Flugel, Big Band Hit, Synth Brass 80, Sax Pad, Trumpet Call.

Strings & Disco Disco Strings, Philly Strings, Cop Show Strings, Mirror Ball Pad, Slow Jam Pad, Staccato Strings, Lush Ensemble, String Machine, Velvet Pad, Closing Credits.

Leads & Talkbox Funk Lead, Chrome Lead, G-Funk Whistle, Talkbox Wah, Robot Cop, Vocoder Cruise, Squeal Lead, Pop Lock Lead, Midnight Cruiser, Disco Solo.

Guitar & FX Chank Guitar, Funk Strum, Wah Guitar, Clean Pluck, Jazz Box, Muted Pick, Siren FX, Radio Dispatch, Disco Ball Bell, Laser Zap.

User presets. Save stores the current sound as a `.fkpreset` file in `Documents\DJBILBOX BEATS\FUNKYKOPS\Presets`. Files in this folder appear in the **User** submenu of the preset menu. Use **Open a .fkpreset file...** to load a preset from another folder. Your sound is also saved inside your FL Studio project.

Production tips

- **G-Funk whine lead.** Load **G-Funk Whistle**: Lead engine, Mono, glide 0.1 and a gentle pitch vibrato. Play it high on the keyboard and push the modulation wheel on long notes for a deeper vibrato. Raise GLIDE for longer slides between notes.
- **Funk bass quack.** On the Synth Bass presets, the sound comes from the filter: a low FILTER, a high ENV and a short DEC give the pluck. Raise RES toward 0.5 for more squelch, as in **Rubber Bass** and **Acid Funk**. Raise **WIDTH** for more sub-octave weight.
- **Clav rhythm parts. Chicken Scratch** and **Stevie Stab** react strongly to velocity (VEL 0.8 to 0.9). Program short sixteenth notes with varied velocities. Raise **TEXTURE** for more pickup bite.
- **Auto-wah in time. Wah Patrol** and **Wah Guitar** use the LFO on the filter. RATE is in Hz: for one wah cycle per beat, set RATE to BPM / 60 (1.67 Hz at 100 BPM). The LFO restarts on each note, so every note gets the same wah shape.
- **Rhodes chords for Hip-Hop and Deep House.** Start from **Rhodes Suitcase** or **Silk Rhodes**. Play harder and raise **BRIGHT** for more bark. **Tremolo EP** uses the LFO on Amp for the classic tremolo. Add CHORUS around 0.3.
- **Disco strings. Disco Strings** and **Philly Strings** stack five detuned saws. **WIDTH** sets the ensemble spread. Double the line one octave up with the same preset for the classic disco string sound.
- **Talking lines with the Talkbox engine.** Automate **VOWEL** in FL Studio to make a line "talk". Or set Destination to Timbre and let the LFO move the vowel, as in **Talkbox Wah**. Keep in mind that Timbre moves all five timbre knobs at once.
- **Delay in time with the beat.** TIME is set in seconds. Quarter note = 60 / BPM. Eighth note = 30 / BPM. Dotted eighth = 45 / BPM. At 96 BPM, an eighth note is 0.31 s.

MIDI, automation and performance

- **Notes.** FUNKYKOPS responds to all MIDI notes (0 to 127) on all MIDI channels, tuned to equal temperament with A4 = 440 Hz. Middle C is MIDI note 60 (C5 in FL Studio).
- **Polyphony.** 16 voices in Poly mode. When you play more than 16 notes, the oldest note is stolen; the lowest and highest held notes are protected. In Mono mode, each new note releases the previous one (its release tail still sounds). Releasing a note does not bring back an earlier held note.
- **Velocity.** Velocity changes the volume according to the VEL slider. On the Rhodes engine, velocity also adds bell "bark".
- **Pitch bend.** ±2 semitones. The MIDI pitch wheel and the on-screen wheel add together.
- **Modulation (CC1).** Adds up to 0.8 to the LFO AMOUNT. The higher of MIDI CC1 and the on-screen wheel is used.
- **Sustain pedal (CC64).** Supported. Aftertouch and other MIDI CCs are ignored. Use host automation for everything else.
- **Automation in FL Studio.** Every parameter in the table can be automated. Move a control on the FUNKYKOPS panel, then choose **Tools > Last tweaked > Create automation clip**. To use a hardware knob, choose **Tools > Last tweaked > Link to controller** and move the knob. Volume, pitch, bend and cutoff changes are smoothed to avoid zipper noise.
- **Latency.** 0 samples. No delay compensation is needed. The plugin reports a 12-second tail to the host so that delay and reverb tails are not cut when you render.
- **CPU.** FUNKYKOPS uses no samples and streams nothing from disk. The CPU load grows with the number of sounding notes. The five-saw Strings engine is the heaviest. Effects at zero are bypassed and use no CPU. Long release times keep voices active longer.
- **Sample rates.** 44.1, 48, 88.2 and 96 kHz.

FAQ

I hear no sound. Check that the FUNKYKOPS channel is routed to a Mixer track and that the channel is not muted. Check VOLUME. With Low Pass selected, a very low FILTER setting can silence the sound. With High Pass selected, a high FILTER setting removes bass presets. Finally, remember that the on-screen keyboard plays very low notes in version 1.0: test with a MIDI keyboard or the Piano roll.

FUNKYKOPS does not appear in FL Studio. FUNKYKOPS is a 64-bit VST3 plugin. Check that `FUNKYKOPS.vst3` is in `C:\Program Files\Common Files\VST3`. In FL Studio, open **Options > Manage plugins** and click **Find installed plugins**. Look for it under **Generators**, manufacturer **DJBILBOX BEATS**.

Can I sing into the Talkbox engine? No. The Talkbox engine is a synthesizer: it shapes saw waves with two vowel formants. FUNKYKOPS has no audio input. Change the vowel with the VOWEL knob, its automation, or the LFO set to Timbre. VOWEL only acts on the Talkbox engine. Some other timbre knobs are also inactive on certain engines; see the engine table in the Parameters section.

The sound gets quieter when I turn REVERB fully down. In version 1.0, the reverb stage also lifts the dry signal by about 6 dB while it is active. At 0 the reverb is bypassed, so the overall level drops. For a dry sound at the usual level, leave REVERB at a very low value such as 0.01, or compensate with VOLUME.

Are the Clavinet and Rhodes engines samples of the original instruments? No. They are synthesized models that evoke those instruments. Clavinet, Rhodes, Moog and Wurlitzer are trademarks of their respective owners. They are used only to describe a sound character. DJBILBOX BEATS is not affiliated with these companies.

GOLDHAWK

Golden, cinematic leads, brass, pads, plucks, bells and basses for epic beats.

Type	Synthesizer (VST3 instrument)
Format	VST3 64-bit (Windows)
Developer	DJBILBOX BEATS
Version	1.0
Factory presets	70
Latency	0 samples

Overview

GOLDHAWK is a 16-voice synthesizer with a royal, cinematic character. It has eight synthesis engines: Hawk Lead (pulse-width-modulated lead), Golden Brass (brass with a self-opening filter), Royal Pad (wide breathing pad), Falcon Pluck (pluck with a pitch dive), Sky Bells (FM bells), Feather Strings (strings with delayed vibrato), Talon Bass (saturated bass with sub) and Eagle Cry (falling, breathy cry for effects). It uses no samples. Every note is generated in real time and responds to the five timbre knobs, the filter and the envelopes.

Each engine runs through a resonant multimode filter with its own envelope, a key-synced LFO and an effects chain: distortion, chorus, stereo ping-pong delay and reverb. A soft limiter at the output keeps the level under control.

GOLDHAWK is built for Trap, Hip-Hop and Drill with an epic or luxurious mood, Afrotrap, cinematic intros and Deep House plucks. Use Hawk Lead for hooks, Golden Brass for fanfares and stabs, Royal Pads for atmospheres, Falcon Plucks and Sky Bells for melodies and arpeggios, Talon Bass for low end and Eagle Cry for transitions.

Quick start in FL Studio

1. In the **Channel rack**, click **+** and choose **GOLDHAWK** (manufacturer **DJBILBOX BEATS**). You can also drag it from **Plugin database > Generators**.
2. The interface opens on **GOLDHAWK - Init Patch**.
3. Click the folder icon at the left of the preset display. Open a bank (for example **Hawk Leads**) and pick a preset. You can also step through all 70 presets with the **<** and **>** arrows.
4. Play notes from your MIDI keyboard, or draw them in the **Piano roll**.
5. Turn **FILTER** to make the sound darker or brighter.
6. Turn **TONE** (below the cassette) to change the character of the engine.
7. Turn **REVERB** to place the sound in a small or large space.
8. Click the disk icon at the right of the preset display to save your own version.

Interface

The window opens at 1000 × 746 pixels. Drag the bottom-right corner to resize it from half size to double size. The proportions stay fixed.

Preset display (top center). From left to right: the folder icon opens the preset menu, **<** loads the previous preset, the center shows the current preset name, **>** loads the next preset, and the disk icon saves a user preset. The preset menu lists the seven factory banks as submenus, a **User** submenu with your saved presets, and an **Open a .ghpreset file...** item.

EXPANSION cassette (top left). Click it to see information about upcoming expansion packs (**EXPANSION VOL.1**) and the link to the DJBILBOX BEATS store.

Small knobs (below the cassette). Two rows of four small knobs. First row: **TONE**, **BODY**, **BRIGHT** and **WIDTH**. Second row: **TEXTURE**, **RES** (filter resonance), **ENV** (filter envelope depth) and **DELAY** (delay level). TONE, BODY, BRIGHT, WIDTH and TEXTURE shape the selected engine.

Logo (top right). The DJBILBOX BEATS name and logo.

Emblem (center). The golden camel-hawk emblem sits between two decorative speaker grilles. The GOLDHAWK title and “By Humpire” are below it.

LFO group (left). RATE and **AMOUNT** set the LFO speed and depth. **PITCH** sits in the same row but is the global transpose of the instrument.

Amplitude envelope (left, below the LFO). Four vertical sliders: **A**, **D**, **S** and **R** (attack, decay, sustain and release of the volume envelope).

ENVELOPE group (right). Four vertical sliders: **A**, **D** and **S** of the filter envelope, and **VEL** (velocity sensitivity of the volume). The filter envelope release uses the same time as its decay. **VOLUME** is to the right of this group.

Effects (right). REVERB, **CHORUS** and **DISTORTION** on one row. Below, the **DELAY** group with **TIME** and **FEEDBACK**. The delay level is the small **DELAY** knob below the cassette.

Center bottom. GLIDE on the left. Four labeled menus in a column: **WAVEFORM** (engine), **LFO SOURCE** (LFO shape), **DESTINATION** (LFO target) and **MONO/POLY** (voice mode). Then the **FILTER** knob (cutoff) and the small **FILTER TYPE** menu.

Wheels (bottom left). The left wheel bends the pitch and springs back to center when you release the mouse. The right wheel is the modulation wheel and stays where you leave it.

On-screen keyboard (bottom). Click a key to audition the sound. It also lights up incoming MIDI notes within its range. In version 1.0 the keyboard covers the lowest part of the MIDI range (keys labeled C-2 to about A0, MIDI notes 0 to 33). Use it to check bass notes. Play melodies and chords from a MIDI keyboard or the Piano roll.

Using the controls. Drag a knob up/down or left/right. Drag a slider vertically. The mouse wheel adjusts in small steps. Double-click to return to the default value. Hold Ctrl or Alt while you drag for velocity-sensitive, finer movement. A small bubble shows the current value while you hover or drag.

Parameters

Names in parentheses are the parameter names shown in FL Studio for automation, when they differ from the panel label.

Parameter	Range	Default	Function
WAVEFORM (Waveform)	Hawk Lead, Golden Brass, Royal Pad, Falcon Pluck, Sky Bells, Feather Strings, Talon Bass, Eagle Cry	Hawk Lead	Selects the synthesis engine. See the engine table below.
LFO SOURCE (LFO Source)	Sine, Triangle, Saw, Square, Sample&Hold	Sine	LFO waveform. Sample&Hold picks a new random value once per cycle.

Parameter	Range	Default	Function
DESTINATION (Destination)	Pitch, Filter, Amp, Timbre	Pitch	Where the LFO goes. Pitch: up to ±1 semitone. Filter: up to ±3 octaves of cutoff. Amp: tremolo, up to 50% volume dip. Timbre: moves all five timbre knobs together, up to ±0.5.
FILTER TYPE (Filter Type)	Low Pass, High Pass, Band Pass	Low Pass	12 dB/octave state-variable filter mode. Band Pass gets a small level boost.
MONO/POLY (Mono/Poly)	Poly, Mono	Poly	Poly: up to 16 notes. Mono: each new note releases the previous one.
RATE (LFO Rate)	0.05 to 20 Hz	4 Hz	LFO speed. The LFO restarts at each new note. It is not tempo-synced.
AMOUNT (LFO Amount)	0 to 1	0	LFO depth. The modulation wheel (or MIDI CC1) adds up to 0.8 on top.
PITCH (Pitch)	-12 to +12 semitones, continuous	0	Global transpose. Values between semitones detune the whole instrument.
GLIDE (Glide)	0 to 2 s	0 (off)	Portamento. Above 0, each new note slides from the pitch of the previous note, in Poly and Mono.
FILTER (Filter)	40 Hz to 18 kHz	9 kHz	Filter cutoff frequency. The cutoff follows the played note at 25% (key tracking).
RES (Resonance)	0 to 1	0.15	Filter resonance (emphasis at the cutoff).
ENV (Filter Env)	0 to 1	0	Filter envelope depth. At 1, the envelope opens the cutoff by up to 5 octaves.
ENVELOPE A (Filter Attack)	0.002 to 3 s	0.002 s	Filter envelope attack time.
ENVELOPE D (Filter Decay)	0.02 to 4 s	0.3 s	Filter envelope decay time. Also sets the filter envelope release after note-off.
ENVELOPE S (Filter Sustain)	0 to 1	0	Filter envelope sustain level.
VEL (Velocity)	0 to 1	0.5	How much note velocity changes the volume. 0 = every note at full level. 1 = full velocity response.
A (Attack)	0.001 to 5 s	0.01 s	Volume envelope attack time.
D (Decay)	0.01 to 5 s	0.4 s	Volume envelope decay time.

Parameter	Range	Default	Function
S (Sustain)	0 to 1	0.7	Volume envelope sustain level.
R (Release)	0.02 to 8 s	0.4 s	Volume envelope release time after note-off.
TONE (Tone)	0 to 1	0.5	Engine timbre control 1. See the engine table.
BODY (Body)	0 to 1	0.5	Engine timbre control 2. See the engine table.
BRIGHT (Brightness)	0 to 1	0.5	Engine timbre control 3. See the engine table.
WIDTH (Width)	0 to 1	0.3	Engine timbre control 4. See the engine table.
TEXTURE (Texture)	0 to 1	0.2	Engine timbre control 5. See the engine table.
VOLUME (Volume)	0 to 1	0.7	Instrument level, before the effects.
DISTORTION (Distortion)	0 to 1	0	Soft saturation. Drive rises up to 15x; the level is compensated.
CHORUS (Chorus)	0 to 1	0	Chorus mix (fixed 0.7 Hz modulation).
DELAY (Delay)	0 to 1	0	Level of the stereo ping-pong delay.
TIME (Delay Time)	0.05 to 1 s	0.35 s	Delay time in seconds.
FEEDBACK (Feedback)	0 to 0.9	0.35	Delay repeats. Higher values give more echoes.
REVERB (Reverb)	0 to 1	0.25	Reverb amount. The room also grows larger as you turn it up.
Pitch bend wheel (Pitch Bend)	-1 to +1	0	Bends all notes by up to ±2 semitones. Adds to the MIDI pitch wheel.
Modulation wheel (Mod Wheel)	0 to 1	0	Adds LFO depth (up to +0.8). The higher of this wheel and MIDI CC1 is used.

Effects order: Distortion, Chorus, Delay, Reverb, then the output soft limiter. Each effect is bypassed while its knob is at zero.

What the timbre knobs do on each engine

Engine	TONE	BODY	BRIGHT	WIDTH	TEXTURE
Hawk Lead: two pulse waves with PWM, plus a sub-octave sine	Pulse-width modulation depth	Detune between the two pulses	Brightness	Pulse-width modulation speed	Added saw layer
Golden Brass: three saws with a filter that opens on each note	Swell time (how slowly the filter opens)	Weight (level)	How far the filter opens	Detune	Breath noise
Royal Pad: five saws and a sub-octave sine, slow breathing filter	No effect	Sub-octave level	Brightness	Detune	Air noise
Falcon Pluck: two saws with a pitch dive and a closing filter	Pitch dive at the attack	Filter closing time	Filter sweep amount	Detune	Attack noise
Sky Bells: FM bell	FM ratio (bell color, from soft to clangorous)	Decay length	FM brightness (metallic shine)	Detune (beating)	Strike noise
Feather Strings: five detuned saws with delayed vibrato	Body resonance pitch	Body resonance level	Brightness	Vibrato depth (starts after 0.2 s)	Bow noise
Talon Bass: three detuned saws, sub-octave sine, saturation	Sub-octave level	Detune	Brightness	Saturation	Noise
Eagle Cry: falling pitch with harmonics and breath	Fall depth and breath color	3rd harmonic	2nd harmonic	Fall speed	Breath level

Factory presets

GOLDHAWK includes 70 factory presets in 7 banks of 10. Loading a factory preset resets every parameter before applying the preset. The pitch bend and modulation wheels are not changed.

Hawk Leads Goldhawk Lead, Soaring, Royal Pulse, Golden Wire, Sky Solo, Hawk Wide, Thermal, Eagle Eye, Golden Stab, Altitude.

Golden Brass Golden Brass, Royal Fanfare, Hawk Stab, Imperial Horns, Soft Flugel, Slow Swell, Low Brass, Gold Rush Hit, Trumpet Call, Cinematic Brass.

Royal Pads Royal Pad, Golden Hour, Palace Air, Rising Hawk, Cloud Throne, Warm Gold, Dream Flight, Dusk Strings, Whisper Wind, Pulse Of Gold.

Falcon Plucks Falcon Pluck, Gold Arp, Dive Zap, Muted Talon, Soft Feather, Pluck Choir, Golden Harp, Royal Koto, Rhythm Gate, Sad Hawk.

Sky Bells & Strings Sky Bell, Golden Chimes, Crystal Crown, Temple Bell, Music Box Gold, Feather Strings, Violin Flight, Slow Rise Strings, Dark Cellos, Pizz Gold.

Talon Bass Talon Bass, Gold Sub, Dirty Talon, Wobble Wing, Industrial Hawk, Slide Bass, Acid Talon, Deep Nest, Bass Stab, Meltdown Gold.

Eagle Cry & FX Eagle Cry, Hawk Screech, Distant Call, Falcon Dive, Wind Rush, Mountain Echo, Golden Whistle, Raptor Siren, Night Hunter, Thunder Wing.

User presets. The disk icon saves the current sound as a `.ghpreset` file in `Documents\DJBILBOX BEATS\GOLDHAWK\Presets`. Files in this folder appear in the **User** submenu of the preset menu. Use **Open a .ghpreset file...** to load a preset from another folder. Your sound is also saved inside your FL Studio project.

Production tips

- **Trap and Drill hooks. Goldhawk Lead** is in Mono with a short glide and a 0.375 s delay (a dotted eighth at 120 BPM). Each note slides briefly from the previous one. Raise **WIDTH** for a faster pulse-width movement, or **TONE** for a deeper one.
- **Cinematic brass.** On the Golden Brass engine, each note opens its own filter. **TONE** sets how slowly it opens: low for punchy stabs (**Hawk Stab**, **Gold Rush Hit**), high for slow swells (**Slow Swell**, **Cinematic Brass**). For intros, layer **Imperial Horns** with **Low Brass**, which is already transposed one octave down.
- **Plucks and arpeggios. Falcon Pluck** and **Gold Arp** work well for Afrotrap and Deep House melodies. **TONE** adds a pitch “dive” at the attack; turn it down for a clean pluck, up for laser zaps (**Dive Zap**). **BODY** sets how long the filter stays open.
- **Bells for dark melodies.** On Sky Bells, **TONE** changes the FM ratio. Most positions give inharmonic, metallic tones. Find a setting that fits your key, then use **BRIGHT** to control the shine and **BODY** for the decay length.
- **Talon Bass as an 808-style bass.** All Talon Bass presets are in Mono and transposed one octave down. For slides, use **Slide Bass** or raise GLIDE: any value above 0 makes every note slide from the previous one. **WIDTH** adds saturation that stays audible on small speakers. Keep **BODY** low for a tight, centered sub.
- **Builds and drops. Rising Hawk** combines a slow attack with a slow filter envelope for builds. Automate **FILTER** over 4 or 8 bars for longer risers. Use **Raptor Siren** or **Falcon Dive** on a single long note before the drop.
- **Delay in time with the beat.** TIME is set in seconds. Quarter note = 60 / BPM. Eighth note = 30 / BPM. Dotted eighth = 45 / BPM. At 140 BPM, a dotted eighth is 0.32 s.
- **Keep chords clean.** The output passes through a soft limiter. Dense pad chords at high VOLUME saturate gently. Lower **VOLUME** on big chords, and raise it on short stabs and plucks, which peak higher than they sound.

MIDI, automation and performance

- **Notes.** GOLDHAWK responds to all MIDI notes (0 to 127) on all MIDI channels, tuned to equal temperament with A4 = 440 Hz. Middle C is MIDI note 60 (C5 in FL Studio).
- **Polyphony.** 16 voices in Poly mode. When you play more than 16 notes, the oldest note is stolen; the lowest and highest held notes are protected. In Mono mode, each new note releases the previous one (its release tail still sounds). Releasing a note does not bring back an earlier held note.
- **Velocity.** Velocity changes the volume according to the VEL slider.
- **Pitch bend.** ±2 semitones. The MIDI pitch wheel and the on-screen wheel add together.
- **Modulation (CC1).** Adds up to 0.8 to the LFO AMOUNT. The higher of MIDI CC1 and the on-screen wheel is used.
- **Sustain pedal (CC64).** Supported. Aftertouch and other MIDI CCs are ignored. Use host automation for everything else.
- **Automation in FL Studio.** Every parameter in the table can be automated. Move a control on the GOLDHAWK panel, then choose **Tools > Last tweaked > Create automation clip**. To use a hardware knob, choose **Tools > Last tweaked > Link to controller** and move the knob. Volume, pitch, bend and cutoff changes are smoothed to avoid zipper noise.

- **Latency.** 0 samples. No delay compensation is needed. The plugin reports a 12-second tail to the host so that delay and reverb tails are not cut when you render.
- **CPU.** GOLDHAWK uses no samples and streams nothing from disk. The CPU load grows with the number of sounding notes. The five-saw Royal Pad and Feather Strings engines are the heaviest. Effects at zero are bypassed and use no CPU. Long release times, common on pads, keep voices active longer.
- **Sample rates.** 44.1, 48, 88.2 and 96 kHz.

FAQ

I hear no sound. Check that the GOLDHAWK channel is routed to a Mixer track and that the channel is not muted. Check VOLUME. With Low Pass selected, a very low FILTER setting can silence the sound. With High Pass selected, a high FILTER setting removes bass presets. Finally, remember that the on-screen keyboard plays very low notes in version 1.0: test with a MIDI keyboard or the Piano roll.

GOLDHAWK does not appear in FL Studio. GOLDHAWK is a 64-bit VST3 plugin. Check that `GOLDHAWK.vst3` is in `C:\Program Files\Common Files\VST3`. In FL Studio, open **Options > Manage plugins** and click **Find installed plugins**. Look for it under **Generators**, manufacturer **DJBILBOX BEATS**.

The Talon Bass presets sound an octave lower than the notes I play. This is intended. Every Talon Bass preset sets **PITCH** to -12 semitones for a deep sub. Set PITCH back to 0 to play at the written pitch.

TONE has no effect on my pad. On the Royal Pad engine, TONE is not used. Use BRIGHT, WIDTH, BODY and TEXTURE to shape the pad. See the engine table in the Parameters section.

The sound gets quieter when I turn REVERB fully down. In version 1.0, the reverb stage also lifts the dry signal by about 6 dB while it is active. At 0 the reverb is bypassed, so the overall level drops. For a dry sound at the usual level, leave REVERB at a very low value such as 0.01, or compensate with VOLUME.

HIGHROLLER GOLD

Casino-luxury synthesizer: golden bells, velvet keys, big band brass, Vegas organ, silk strings and gold 808s from eight synthesis engines.

Type	Synthesizer (VST3 instrument)
Format	VST3 64-bit (Windows)
Developer	DJBILBOX BEATS
Version	1.0
Factory presets	70
Latency	0 samples

Overview

HIGHROLLER GOLD is a synthesizer by Humpire / DJBILBOX BEATS with a high-stakes, gold-and-velvet character. It uses no samples. Every sound is generated in real time by one of eight engines: Gold Bell (FM bell), Velvet Keys (electric piano with tine and bell partials), Big Band Brass (three detuned saws with a swelling filter), Vegas Organ (drawbar organ with percussion and key click), Roulette Mallets (vibraphone-style bars with tremolo), Silk Strings (five-saw ensemble with slow drift), Gold Sub (sine sub with pitch “knock” and saturation) and Cigar Smoke (detuned saw pad with a breathing filter).

Five character controls (TONE, BODY, BRIGHT, WIDTH, TEXTURE) reshape each engine. The sound then goes through a multimode filter with its own envelope, an amplitude envelope, an LFO and a master effects chain: distortion, chorus, stereo delay and reverb. The result is smooth, expensive-sounding and slightly smoky: jackpot bells, late-night lounge keys, brass fanfares and deep subs.

HIGHROLLER GOLD targets luxury and mafioso Hip-Hop, Trap bells and 808s, G-Funk and West Coast keys, Boom Bap and lo-fi chords, R&B and Deep House pads and organs. It plays up to 16 notes at once. The Gold Sub & Bass bank loads in Mono mode for clean bass lines.

Quick start in FL Studio

1. In the **Channel rack**, click **+** and choose **HIGHROLLER_GOLD** (vendor **DJBILBOX BEATS**). You can also drag it from the **Plugin database > Generators** browser.
2. The plugin opens on **HIGHROLLER GOLD - Init Patch**: the Gold Bell engine in Poly mode.
3. Click the **folder icon** at the left of the preset display. Pick a bank, then a preset. You can also step through all 70 presets with the **<** and **>** arrows.
4. Play from your MIDI keyboard or draw notes in the **Piano roll**. Keys, bells and pads sit well between C4 and C6 in FL Studio. Bass presets sit between C2 and C4.
5. Shape the sound with the three main controls:
   - **BRIGHT**: the brightness of the engine (FM depth, key “bark”, brass opening, upper drawbars...).
   - **BODY**: the length or weight of the sound (ring time for bells, keys and mallets, sub level for pads and basses).
   - **FILTER**: the cutoff of the filter. Turn it down for a darker, warmer, “tape” tone.
6. Adjust **VOLUME** to set the level, then click the **disk icon** to save your own version.

Interface

The window opens at 1000 × 746 pixels. Drag the bottom-right corner to resize it from 50% to 200%. The proportions stay fixed.

Preset display (top center). From left to right: the folder icon (preset menu), the **<** arrow, the current preset name, the **>** arrow and the disk icon (save). The arrows wrap around from the last preset to the first.

EXPANSION cassette (top left). Clicking the EXPANSION plate under the cassette opens an information message about upcoming expansion packs, with the address of the DJBILBOX BEATS store (djbilboxbeats.gumroad.com). No expansion content is included with version 1.0.

Logo (top right). The DJBILBOXBEATS name and the DB logo.

Character row (under the preset display). Five small knobs: **TONE, BODY, BRIGHT, WIDTH, TEXTURE**. They are the character controls of the selected engine.

Center. The camel mascot in a fedora, framed by two gold-dotted speaker grilles. The grilles are decorative. Below them: the HIGHROLLER GOLD title and “By Humpire”.

LFO section (left). RATE and **AMOUNT** set the LFO speed and depth. **PITCH**, also under this header, is the global transposition (-12 to +12 semitones). It is not an LFO control.

Amplitude envelope (left). ATTACK, DECAY, SUSTAIN, RELEASE sliders.

Filter controls (left of the title). RES (resonance) and **ENV** (filter envelope depth), two small knobs. **GLIDE** sits below them.

Right of the title. VOLUME and, below it, **FILTER** (cutoff).

Filter envelope (top right). A, D, S, R sliders for the filter envelope, then **VELOCITY** (velocity sensitivity of the level).

Effects (right). First row: **REVERB, CHORUS, DISTORTION**. Second row: **DELAY** (delay level), **TIME** and **FEEDBACK**.

Menu row. Five drop-down menus, from left to right: **Waveform** (engine), **LFO Source** (LFO shape), **Destination** (LFO target), **Filter Type** and **Mono/Poly**. The menus have no title. Each one shows its current value.

Wheels (bottom left). Pitch Bend springs back to the center when you release it. **Modulation** stays where you leave it.

On-screen keyboard (bottom). Click a key to play it. Velocity follows the click position: click near the bottom of a key for a louder note. The keyboard starts at the lowest MIDI note (labeled C-2, which is C0 in FL Studio) and does not scroll. It covers MIDI notes 0 to 31, which are very low. Use a MIDI controller or the Piano roll for normal playing.

Mouse gestures. Drag a knob up or right to increase it. Drag sliders and wheels vertically. Double-click any knob or slider to return it to its default value. The mouse wheel also adjusts controls. A value bubble appears while you drag or hover.

Parameters

Signal flow: engine, then filter, then amplitude envelope (per voice). The voices are summed, then processed by Distortion, Chorus, Delay and Reverb. A DC blocker and a soft limiter protect the output.

HIGHROLLER GOLD exposes 35 automatable parameters. The host name is shown in parentheses when it differs from the panel label.

Parameter	Range	Default	Function
Waveform (menu)	Gold Bell, Velvet Keys, Big Band Brass, Vegas Organ, Roulette Mallets, Silk Strings, Gold Sub, Cigar Smoke	Gold Bell	Selects the synthesis engine.
TONE (Tone)	0 to 1	0.5	Character control 1. Its effect depends on the engine (see the table below).
BODY (Body)	0 to 1	0.5	Character control 2. Engine-dependent.
BRIGHT (Brightness)	0 to 1	0.5	Character control 3. Engine-dependent.
WIDTH (Width)	0 to 1	0.3	Character control 4. Engine-dependent (detune, percussion, tremolo depth...). It is not a stereo width control.
TEXTURE (Texture)	0 to 1	0.2	Character control 5. Engine-dependent (attack noise, key click, saturation...).
FILTER (Filter)	40 to 18,000 Hz	9,000 Hz	Filter cutoff. The cutoff follows the keyboard by 25% around middle C.
RES (Resonance)	0 to 1	0.15	Filter resonance.
Filter Type (menu)	Low Pass, High Pass, Band Pass	Low Pass	Filter mode (12 dB per octave).
ENV (Filter Env)	0 to 1	0	Filter envelope depth. At 1 the envelope opens the cutoff up to 5 octaves. At 0 the filter envelope has no effect.
A (Filter Attack)	0.002 to 3 s	0.002 s	Filter envelope attack time.
D (Filter Decay)	0.02 to 4 s	0.3 s	Filter envelope decay time.
S (Filter Sustain)	0 to 1	0	Filter envelope sustain level.
R (Filter Release)	0.02 to 4 s	0.3 s	Filter envelope release time.
VELOCITY (Velocity)	0 to 1	0.5	How much note velocity changes the level. 0: all notes at the same level. 1: full velocity response.
ATTACK (Attack)	0.001 to 5 s	0.01 s	Amplitude envelope attack time.
DECAY (Decay)	0.01 to 5 s	0.4 s	Amplitude envelope decay time.

Parameter	Range	Default	Function
SUSTAIN (Sustain)	0 to 1	0.7	Amplitude envelope sustain level.
RELEASE (Release)	0.02 to 8 s	0.4 s	Amplitude envelope release time.
VOLUME (Volume)	0 to 1	0.7	Level of each voice, before the effects.
GLIDE (Glide)	0 to 2	0	Portamento. The value is roughly the slide time in seconds. Each new note slides from the pitch of the previous note, in Mono and in Poly mode. 0 turns glide off.
PITCH (Pitch)	-12 to +12 semitones	0	Global transposition. The control is continuous: use whole values to stay in tune.
RATE (LFO Rate)	0.05 to 20 Hz	4 Hz	LFO speed. Not tempo-synced.
AMOUNT (LFO Amount)	0 to 1	0	LFO depth. The mod wheel adds to this value.
LFO Source (menu)	Sine, Triangle, Saw, Square, Sample&Hold	Sine	LFO shape. The LFO restarts at every new note. Sample&Hold picks a new random value once per cycle.
Destination (menu)	Pitch, Filter, Amp, Timbre	Pitch	LFO target. Pitch: up to +/-1 semitone (vibrato). Filter: up to +/-3 octaves. Amp: up to 50% level dip (tremolo). Timbre: moves all five character controls by up to +/-0.5.
Mono/Poly (menu)	Poly, Mono	Poly	Poly: up to 16 notes at once. Mono: one note at a time, last-note priority.
DISTORTION (Distortion)	0 to 1	0	Saturation on the summed output, with automatic level compensation.
CHORUS (Chorus)	0 to 1	0	Stereo chorus mix (slow, fixed rate).
DELAY (Delay)	0 to 1	0	Level of the stereo delay. The repeats cross between the left and right channels.
TIME (Delay Time)	0.05 to 1.0 s	0.35 s	Delay time in seconds. Not tempo-synced: for a quarter note, use 60 / BPM (0.5 s at 120 BPM); halve it for an eighth note.

Parameter	Range	Default	Function
FEEDBACK (Feedback)	0 to 0.9	0.35	Number of delay repeats (0 to 90% feedback).
REVERB (Reverb)	0 to 1	0.25	Reverb amount. Higher values also enlarge the room.
Pitch Bend wheel (Pitch Bend)	-1 to +1	0	On-screen pitch bend, +/-2 semitones. Adds to MIDI pitch bend.
Modulation wheel (Mod Wheel)	0 to 1	0	On-screen mod wheel. Adds up to +0.8 to the LFO AMOUNT.

Character controls by engine

Engine	TONE	BODY	BRIGHT	WIDTH	TEXTURE
Gold Bell	FM ratio (1.4 to 4.5): from warm bell to metallic chime	Ring time of the bell	FM depth (strike brightness)	Slight detune of a second tone (beating)	Strike noise
Velvet Keys	No effect	Decay length of the tine and bell partials	“Bark”: level of the upper partials, stronger on hard hits	No effect	Hammer noise at the attack
Big Band Brass	Swell time of the built-in filter (fast to slow)	Output weight	How far the built-in filter opens	Detune of the three saws	Breath noise
Vegas Organ	Sub-octave drawbar (16′)	Octave drawbar (4′)	Upper drawbars (2 2/3′, 2′, 1 1/3′)	Percussion (short decaying octave)	Key click
Roulette Mallets	Mallet attack noise	Decay length	Level of the upper partials	Tremolo depth (up to 60%)	Tremolo speed (2 to 9 Hz)
Silk Strings	Body resonance frequency (350 to 850 Hz)	Body resonance level	Built-in low-pass opening	Detune and slow pitch drift	Drift speed (0.5 to 2 Hz)
Gold Sub	Pitch “knock” at the attack	Adds the octave harmonic	Attack click	No effect	Saturation
Cigar Smoke	No effect	Sub-octave level	Built-in low-pass opening (with a slow “breathing” movement)	Detune of the five saws	Breath noise

The Gold Bell, Velvet Keys and Roulette Mallets engines decay naturally, like the real instruments. Holding a key does not sustain them forever. Raise BODY for a longer ring.

Factory presets

HIGHROLLER GOLD includes 70 factory presets in 7 banks of 10. All presets load in Poly mode, except the Gold Sub & Bass bank and Cigar Lounge Horn, which load in Mono mode.

Gold Bells & Jackpot: Jackpot Bell, Gold Coin, Slot Machine, Casino Chimes, Velvet Box, Crystal Chandelier, Low Gong, Ring Of Fortune, Lucky Seven, Winner Echo.

Velvet Keys: Velvet Rhodes, High Stakes Keys, Bark EP, Lounge Keys, Tremolo Velvet, After Dark EP, Wurly Bite, Gilded Bell EP, Penthouse Chords, Tape Velvet.

Big Band Brass: Big Band Brass, High Roller Horns, Brass Stab, Soft Flugel, Slow Swell, Vegas Fanfare, Muted Horn, Sax Section, Disco Hit, Cigar Lounge Horn.

Vegas Organ: Vegas Organ, Smoky Lounge, Full Drawbars, Gospel Gold, Casino Floor, Perc Organ, Soft Flute Stop, Rotary Dream, Tape Organ, Golden Swell.

Mallets & Strings: Roulette Vibes, Gold Marimba, Chip Stack, Slow Motor Vibes, Dealer's Hand (Roulette Mallets engine); Silk Strings, Velvet Strings, Slow Rise Strings, Dark Cellos, Pizz Casino (Silk Strings engine).

Gold Sub & Bass: Gold 808, Long Slide Gold, Dirty Gold, Boom Bap Sub, Hard Knock, Deep Vault, Short Gold, Smooth Sub, Pitch Drop Gold, Rumble Royale.

Cigar Smoke Pads: Cigar Smoke, Smoky Backroom, Velvet Curtain, Golden Haze, High Roller Rise, Midnight Casino, Whisper Table, Vault Drone, Pulse Of Luck, Final Hand.

Some presets are transposed with PITCH. One octave up: Gold Coin, Slot Machine, Casino Chimes, Crystal Chandelier, Gilded Bell EP, Chip Stack, Golden Haze. One octave down: Low Gong, After Dark EP, Dark Cellos, Deep Vault, Smoky Backroom, Vault Drone.

Loading a preset first resets every parameter to its default value, then applies the preset. The two wheels are not changed.

User presets. Click the disk icon to save the current sound as a `.hgpreset` file. The default folder is `Documents\DJBILBOX BEATS\HIGHROLLER_GOLD\Presets`. Files in this folder appear in the **User** submenu of the preset menu. Use **Open a .hgpreset file...** in the same menu to load a file from another location. The preset display shows the file name.

The host also sees the 70 factory presets as a program list. Your project saves the full state of the plugin, including the preset name.

Rhodes and Wurlitzer are trademarks of their respective owners. Preset names that refer to them only describe a sound style. DJBILBOX BEATS is not affiliated with the trademark owners.

Production tips

1. **Trap bells and melodies.** Start from Jackpot Bell or Gold Coin. TONE changes the FM ratio, so small moves change the bell color a lot: settle on a value that sounds in tune with your track. Raise BRIGHT for a harder strike. Set TIME to 60 / BPM x 0.75 for a dotted-eighth echo.
2. **Luxury Hip-Hop keys.** Use the Velvet Keys bank (Velvet Rhodes, Penthouse Chords). The "bark" responds to velocity, so set VELOCITY around 0.7 to 0.9 and vary note velocities in the Piano roll. Add CHORUS between 0.3 and 0.45 for a wide, glossy electric piano.
3. **G-Funk and West Coast.** Pair a Vegas Organ preset (Smoky Lounge, Perc Organ) with a Gold Sub bass. On the organ, WIDTH adds the percussion and TEXTURE the key click. For a whining lead, set Mono mode with GLIDE around 0.1.
4. **Boom Bap and lo-fi.** Load Tape Velvet or Tape Organ. Pull FILTER down to 2,000 to 3,500 Hz, add a touch of DISTORTION (0.1 to 0.2) and use REVERB sparingly so the chords stay dry and dusty.
5. **808s and subs.** Gold 808 and Hard Knock are punchy 808-style subs; Short Gold is a short, plucked version. TONE sets the knock and TEXTURE the saturation, which helps the bass on small speakers. Long Slide Gold has glide set for 808 slides: every new note slides from the previous one.

6. **Brass stabs and fanfares.** Brass Stab and Disco Hit are short and punchy. For slow swells, raise TONE (slower filter swell) and ATTACK, as in Slow Swell. Short presets are quieter by nature: raise VOLUME.
7. **R&B and Deep House pads.** The Cigar Smoke Pads bank has long attacks and releases. For a moving pad, set Destination to Filter with a slow RATE (0.1 to 0.3 Hz), as in Midnight Casino. High Roller Rise uses the filter envelope (ENV) for a slow opening.
8. **Rotary-style organ.** Rotary Dream uses the Timbre destination with CHORUS to suggest a spinning speaker. Move RATE between 1 and 6 Hz to switch between slow and fast movement.

MIDI, automation and performance

Notes and channels. HIGHROLLER GOLD responds to all 128 MIDI notes on any MIDI channel. Velocity controls the level according to the VELOCITY slider. On Velvet Keys, velocity also controls the "bark".

Polyphony. Up to 16 voices in Poly mode. When you play a 17th note, a voice is taken over and cut immediately. In Mono mode, each new note releases the previous one (last-note priority); a long RELEASE can briefly overlap the two notes. Releasing the newest note does not bring back an earlier note that you still hold.

Pitch bend. MIDI pitch bend covers +/-2 semitones. The on-screen Pitch Bend wheel adds another +/-2 semitones.

Mod wheel (CC1). CC1 raises the LFO depth, like the on-screen Modulation wheel. When both are used, the higher value wins. CC1 acts on the notes that are sounding when the message arrives, so move the wheel while you play. For repeatable results in a song, automate the on-screen **Mod Wheel** parameter instead.

Other MIDI. The sustain pedal (CC64) is supported. In Mono mode, avoid the pedal: notes held by the pedal keep sounding together. All Notes Off and All Sound Off messages stop the sound. Aftertouch is ignored.

Automation in FL Studio. All 35 parameters can be automated. Move a control in the plugin, then use **Tools > Last tweaked > Create automation clip**. Use **Tools > Last tweaked > Link to controller** to assign a hardware knob. You can also pick parameters from the plugin wrapper menu (**Browse parameters**). Level, pitch, bend, filter, distortion and delay changes are smoothed to avoid clicks.

Latency and tail. The plugin adds no latency (0 samples), so no delay compensation is needed. It reports a 12-second tail to the host so reverb and delay trails are not cut when rendering.

CPU. HIGHROLLER GOLD is lightweight. Effects set to zero are bypassed and cost no CPU. The five-saw engines (Silk Strings, Cigar Smoke) use a little more CPU than the others. Short release times keep the voice count low. The plugin works at 44.1, 48, 88.2 and 96 kHz.

FAQ

HIGHROLLER GOLD does not appear in FL Studio. Open **Options > Manage plugins**, check that `C:\Program Files\Common Files\VST3` is in the search paths, then click **Find installed plugins**. The plugin is listed as **HIGHROLLER_GOLD** under **DJBILBOX BEATS**. It is an instrument: load it in the Channel rack, not in a Mixer insert slot.

I hear nothing, or almost nothing, when I click the on-screen keyboard. The on-screen keyboard covers the lowest MIDI notes (0 to 31), and most of them are below what speakers can reproduce. Play from a MIDI keyboard or the Piano roll around C4 to C6 (FL Studio naming). Also check VOLUME, and check that FILTER is not closed with Filter Type set to Low Pass.

My bell or electric piano dies out while I still hold the key. Gold Bell, Velvet Keys and Roulette Mallets decay on their own, like real bells and pianos. Raise BODY for a longer ring, and raise SUSTAIN and RELEASE if the amplitude envelope also cuts the note.

My chord only plays one note. The Mono/Poly menu is set to Mono, which is the case for the Gold Sub & Bass presets and Cigar Lounge Horn. Switch it to Poly.

The sound gets squashed when I play loud chords. A soft limiter protects the output. Lower VOLUME, DISTORTION or the number of notes, and use the FL Studio mixer to set the final level.

MIDNIGHT REELS

Film noir and late-night jazz: vibes, electric piano, muted horns, upright bass, organ, strings and celesta.

Type	Synthesizer (VST3 instrument)
Format	VST3 64-bit (Windows)
Developer	DJBILBOX BEATS
Version	1.0
Factory presets	70
Latency	0 samples

Overview

MIDNIGHT REELS is a 16-voice synthesizer for the sound of a smoky jazz club and classic film noir scores. It has eight synthesis engines: Vibes (vibraphone with a motor tremolo), Reel EP (electric piano with a velocity “bark”), Muted Trumpet, Tenor Sax, Upright Bass (plucked string model), Organ (drawbar-style), Noir Strings (detuned string ensemble with tape flutter) and Celesta. It uses no samples. Every note is generated in real time and responds to five timbre knobs, a filter and two envelopes.

Each voice runs through a resonant multimode filter with its own envelope and a per-voice LFO. The output then goes through an effects chain: distortion, chorus, stereo delay and reverb. A soft limiter at the output keeps the level under control.

MIDNIGHT REELS is built for Jazz-Hop and lo-fi Hip-Hop, Boom Bap, Soul and G-Funk (organ and electric piano), dark Trap and Drill (noir strings, celesta, music box) and Deep House chords. It also fits cinematic intros, detective themes and podcast beds. Use the Reel EP and Organ banks for chords, the Trumpet & Sax bank for hooks, the Upright Bass bank for walking lines, and the Vibes, Celesta and Strings banks for atmosphere.

Quick start in FL Studio

1. In the **Channel rack**, click **+** and choose **MIDNIGHT_REELS** (manufacturer **DJBILBOX BEATS**). You can also drag it from **Plugin database > Generators**.
2. The interface opens on **MIDNIGHT REELS - Init Patch** (Vibes engine).
3. Click the folder icon at the left of the preset display. Open a bank (for example **Reel EP**) and pick a preset. You can also step through all 70 presets with the **<** and **>** arrows.
4. Play notes from your MIDI keyboard, or draw them in the **Piano roll**.
5. Turn **FILTER** to make the sound darker or brighter.
6. Turn **BODY** (center, small knob) to change the body of the engine. On Vibes, Reel EP and Celesta it sets how long the notes ring.
7. Turn **REVERB** to place the sound in a small room or a large hall.
8. Click the disk icon at the right of the preset display to save your own version.

Interface

The window opens at 1000 × 746 pixels. Drag the bottom-right corner to resize it from half size (500 × 373) to double size (2000 × 1492). The proportions stay fixed.

Preset display (top center). From left to right: the folder icon opens the preset menu, **<** loads the previous preset, the center shows the current preset name, **>** loads the next preset, and the disk icon saves a user preset. The arrows wrap around from the last preset to the first. The preset menu lists the seven factory banks as submenus, a **User** submenu with your saved presets, and an **Open a .mrpreset file...** item.

EXPANSION cassette (top left). Click it to see information about upcoming expansion packs (**EXPANSION VOL.1**) and the link to the DJBILBOX BEATS store. Version 1.0 contains no expansion content.

Logo (top right). The DJBILBOX BEATS name and logo.

Emblem (center). The camel detective, in hat and trench coat, sits between two decorative speaker grilles. The MIDNIGHT REELS title and “By Humpire” are below it.

Filter envelope, VEL and VOLUME (right, below the logo). Five vertical sliders: **A**, **D** and **R** are the attack, decay and release of the filter envelope. **VEL** sets the velocity sensitivity of the volume. **VOLUME** is the instrument level.

LFO and transpose (left, below the cassette). RATE and **AMOUNT** set the LFO speed and depth. **PITCH** sits in the same row but is the global transpose of the instrument, not an LFO control.

Volume envelope (left, below RATE/AMOUNT/PITCH). Four vertical sliders: **ATTACK**, **DECAY**, **SUSTAIN** and **RELEASE**.

Center controls. On the left, four small knobs in a square: **TONE**, **BODY**, **BRIGHT** and **WIDTH**. In the middle, **GLIDE** and **FILTER** (cutoff). On the right, three small knobs: **TEXTURE**, **RES** (filter resonance) and **ENV** (filter envelope depth). TONE, BODY, BRIGHT, WIDTH and TEXTURE shape the selected engine.

Effects (right). REVERB, **CHORUS** and **DISTORTION** on the upper row. **DELAY** (level), **TIME** and **FEEDBACK** on the lower row.

Menu row. Five drop-down menus across the panel. Each one shows its current value. From left to right: the engine (**Waveform**, default “Vibes”), the LFO shape (**LFO Source**, default “Sine”), the LFO target (**Destination**, default “Pitch”), the filter mode (**Filter Type**, default “Low Pass”) and the voice mode (**Mono/Poly**, default “Poly”).

Wheels (bottom left). The left wheel bends the pitch and springs back to center when you release the mouse. The right wheel is the modulation wheel and stays where you leave it.

On-screen keyboard (bottom). Click a key to audition the sound. It also lights up incoming MIDI notes within its range. In version 1.0 the keyboard covers the lowest part of the MIDI range (keys labeled C-2 to E1, MIDI notes 0 to 40). Use it to check bass notes. Play melodies and chords from a MIDI keyboard or the Piano roll.

Using the controls. Drag a knob up/down or left/right. Drag a slider vertically. The mouse wheel adjusts in small steps. Double-click to return to the default value. Hold Ctrl or Alt while you drag for velocity-sensitive movement. A small bubble shows the current value while you hover or drag.

Parameters

Names in parentheses are the parameter names shown in FL Studio for automation, when they differ from the panel label.

Parameter	Range	Default	Function
Engine menu (Waveform)	Vibes, Reel EP, Muted Trumpet, Tenor Sax, Upright Bass, Organ, Noir Strings, Celesta	Vibes	Selects the synthesis engine. See the engine table below.

Parameter	Range	Default	Function
LFO shape menu (LFO Source)	Sine, Triangle, Saw, Square, Sample&Hold	Sine	LFO waveform. Sample&Hold picks a new random value once per cycle.
LFO target menu (Destination)	Pitch, Filter, Amp, Timbre	Pitch	Where the LFO goes. Pitch: up to ±1 semitone. Filter: up to ±3 octaves of cutoff. Amp: tremolo, up to 50% volume dip. Timbre: moves all five timbre knobs together, up to ±0.5.
Filter mode menu (Filter Type)	Low Pass, High Pass, Band Pass	Low Pass	12 dB/octave state-variable filter mode. Band Pass gets a small level boost.
Voice mode menu (Mono/Poly)	Poly, Mono	Poly	Poly: up to 16 notes. Mono: each new note releases the previous one.
RATE (LFO Rate)	0.05 to 20 Hz	4 Hz	LFO speed. The LFO restarts at each new note. It is not tempo-synced.
AMOUNT (LFO Amount)	0 to 1	0	LFO depth. The modulation wheel (or MIDI CC1) adds up to 0.8 on top.
PITCH (Pitch)	-12 to +12 semitones, continuous	0	Global transpose. Values between semitones detune the whole instrument.
GLIDE (Glide)	0 to 2 s	0 (off)	Portamento. Above 0, each new note slides from the pitch of the previous note, in Poly and Mono.
FILTER (Filter)	40 Hz to 18 kHz	9 kHz	Filter cutoff frequency. The cutoff follows the played note at 25% (key tracking).
RES (Resonance)	0 to 1	0.15	Filter resonance (emphasis at the cutoff).
ENV (Filter Env)	0 to 1	0	Filter envelope depth. At 1, the envelope opens the cutoff by up to 5 octaves.
A (Filter Attack)	0.002 to 3 s	0.002 s	Filter envelope attack time.
D (Filter Decay)	0.02 to 4 s	0.3 s	Filter envelope decay time. The filter envelope has no sustain stage: it always decays back to zero.
R (Filter Release)	0.02 to 4 s	0.3 s	Filter envelope release time, used when you release a note before the decay has finished.

Parameter	Range	Default	Function
VEL (Velocity)	0 to 1	0.5	How much note velocity changes the volume. 0 = every note at full level. 1 = full velocity response.
ATTACK (Attack)	0.001 to 5 s	0.01 s	Volume envelope attack time.
DECAY (Decay)	0.01 to 5 s	0.4 s	Volume envelope decay time.
SUSTAIN (Sustain)	0 to 1	0.7	Volume envelope sustain level.
RELEASE (Release)	0.02 to 8 s	0.4 s	Volume envelope release time after note-off.
TONE (Tone)	0 to 1	0.5	Engine timbre control 1. See the engine table.
BODY (Body)	0 to 1	0.5	Engine timbre control 2. See the engine table.
BRIGHT (Brightness)	0 to 1	0.5	Engine timbre control 3. See the engine table.
WIDTH (Width / Vibrato)	0 to 1	0.3	Engine timbre control 4. See the engine table.
TEXTURE (Texture)	0 to 1	0.2	Engine timbre control 5. See the engine table.
VOLUME (Volume)	0 to 1	0.7	Instrument level, before the effects.
DISTORTION (Distortion)	0 to 1	0	Soft saturation. Drive rises up to 15x; the level is compensated.
CHORUS (Chorus)	0 to 1	0	Chorus mix (fixed 0.7 Hz modulation).
DELAY (Delay)	0 to 1	0	Level of the stereo cross-feedback (ping-pong style) delay.
TIME (Delay Time)	0.05 to 1 s	0.35 s	Delay time in seconds. Changes glide smoothly, like a tape delay.
FEEDBACK (Feedback)	0 to 0.9	0.35	Delay repeats. Higher values give more echoes.
REVERB (Reverb)	0 to 1	0.25	Reverb amount. The room also grows larger as you turn it up.
Pitch bend wheel (Pitch Bend)	-1 to +1	0	Bends all notes by up to ±2 semitones. Adds to the MIDI pitch wheel.
Modulation wheel (Mod Wheel)	0 to 1	0	Adds LFO depth (up to +0.8). The higher of this wheel and MIDI CC1 is used.

Effects order: Distortion, Chorus, Delay, Reverb, then the output soft limiter. Each effect is bypassed while its knob is at zero.

What the timbre knobs do on each engine

Engine	TONE	BODY	BRIGHT	WIDTH	TEXTURE
Vibes: three bar partials, mallet noise and a motor tremolo	Mallet hardness (attack noise)	Ring length (0.6 to 3.6 s)	Upper partials	Motor tremolo depth (up to 60%)	Motor speed (2 to 9 Hz)
Reel EP: tine partials with a bell overtone	No effect	Decay length	Bark: upper partials, stronger at high velocity	No effect	Key noise at the attack
Muted Trumpet: two saws through a resonant mute	Mute resonance pitch (600 to 2400 Hz)	Mute resonance sharpness and level	Brightness of the open horn layer	Detune between the two saws	Breath at the start of the note
Tenor Sax: reed wave with two formants and delayed vibrato	Formant position (dark to open)	Formant level	Brightness	Vibrato depth (5 Hz; starts after 0.25 s and fades in)	Breath noise
Upright Bass: plucked string model with body resonance	Pluck brightness and finger noise	Sustain	String brightness	No effect	Body resonance (low thump around 95 Hz)
Organ: six drawbar-style harmonics, percussion and key click	Sub drawbar (one octave below)	Octave drawbar (2nd harmonic)	Upper drawbars (3rd, 4th and 6th harmonics)	Percussion (fast-decaying 2nd harmonic)	Key click
Noir Strings: five detuned saws with tape flutter and body resonance	Body resonance pitch	Body resonance level	Brightness	Detune spread and flutter depth	Flutter speed (0.5 to 2 Hz)
Celesta: three bell partials and mallet noise	Mallet hardness (attack noise)	Ring length (0.35 to 2.15 s)	Upper partials	No effect	No effect

On the Organ engine, the main (fundamental) drawbar is fixed. TONE, BODY and BRIGHT add the other drawbars around it.

Factory presets

MIDNIGHT REELS includes 70 factory presets in 7 banks of 10. Loading a factory preset resets every parameter before applying the preset. The pitch bend and modulation wheels are not changed. Most banks use the engine of the same name; exceptions are noted.

Vibes & Mallets (Vibes engine) Midnight Vibes, Smoke Vibes, Hard Mallet Vibes, Slow Motor, Fast Motor, Rainy Window, Low Vibes, Dry Vibes, Ghost Mallets, Vibes Delay.

Reel EP (Reel EP engine) Reel Electric Piano, Noir Keys, Bark EP, Lounge EP, Tremolo EP, Tape EP, After Hours, Wurly Bite, Bell EP, Detective Chords.

Trumpet & Sax (Muted Trumpet and Tenor Sax engines) Muted Trumpet, Harmon Mute, Wah Trumpet, Lonely Horn, Trumpet Stab, Tenor Sax, Smoky Sax, Breathy Sax, Late Night Sax, Growl Sax.

Upright Bass (Upright Bass engine; Bowed Cello uses Noir Strings) Walking Upright, Slap Upright, Deep Pocket, Muted Bass, Jazz Club Bass, Bowed Cello, Detective Walk, Tape Bass, Fat Pluck, Sub Noir.

Organ & Lounge (Organ engine) Jazz Organ, Smoky Lounge, Full Drawbars, Gospel Hammond, Church Hall, Perc Organ, Tape Organ, Soft Flute Stop, Rotary Dream, Noir Swell.

Noir Strings (Noir Strings engine) Noir Strings, Tape Strings, Slow Swell, Dark Cellos, High Violins, Pizzicato Noir, Film Score Pad, Wobble Reel, Suspense, Rooftop Strings.

Celesta & Reels (Celesta engine; Projector Click uses Organ, Reel To Reel uses Reel EP, Closing Credits uses Noir Strings) Music Box Noir, Celesta Dream, Glass Bells, Lullaby Reel, Toy Piano, Haunted Box, Low Celesta, Projector Click, Reel To Reel, Closing Credits.

User presets. The disk icon saves the current sound as a `.mrpreset` file in `Documents\DJBILBOX BEATS\MIDNIGHT_REELS\Presets`. Files in this folder appear in the **User** submenu of the preset menu. Use **Open a .mrpreset file...** to load a preset from another folder. Your sound is also saved inside your FL Studio project.

Production tips

- **Lo-fi and Jazz-Hop keys.** Start from **Tape EP**: the filter is already at 3 kHz and the chorus adds a light wobble. Raise **VEL** so soft notes stay mellow and hard notes bark. On Reel EP, **BRIGHT** sets the bark, and it grows with velocity. Add a touch of **DISTORTION** (0.1 to 0.2) for warmth, as in **Wurly Bite**.
- **Walking bass for Boom Bap.** All Upright Bass presets are in Mono. **TONE** sets the pluck attack, **BODY** the sustain, **TEXTURE** the low body thump around 95 Hz. Keep **DECAY** short for staccato lines. If the bass sits too low in the mix, raise **VOLUME**: plucked notes peak high but sound quieter than pads.
- **Vibes with or without motor.** On the Vibes engine, **WIDTH** is the motor tremolo depth and **TEXTURE** its speed. Set WIDTH to 0 for a static vibraphone. **Slow Motor** and **Fast Motor** show a deep slow motor and a deep fast motor.
- **Horn hooks.** For sax solos, use **Late Night Sax** (Mono with a short glide). The vibrato starts after a quarter second, so short notes stay straight and long notes sing. Raise **WIDTH** for a wider vibrato. **Wah Trumpet** uses the LFO on **Timbre**: slow the **RATE** down for a hand-wah effect in time with your groove.
- **Organ for Soul, Gospel and G-Funk. TONE** adds the sub drawbar, **BODY** the octave, **BRIGHT** the upper drawbars, **WIDTH** the percussion and **TEXTURE** the key click. Add **CHORUS** (0.4 to 0.6) for movement and a little **DISTORTION** for grit.
- **Film noir pads and builds.** On Noir Strings, **WIDTH** sets the detune and tape flutter depth, **TEXTURE** the flutter speed. For a slow reveal, copy the idea of **Suspense**: low **FILTER** (900 Hz), **ENV** at 0.4 and a long filter **A**. Automate **FILTER** over 8 bars for longer risers.
- **Dark Trap melodies. Music Box Noir**, **Haunted Box** and **Glass Bells** cut through dense 808s. Music Box Noir and Glass Bells are already transposed one octave up (**PITCH** +12). Keep **REVERB** around 0.5, and lower **FILTER** if they get too sharp.
- **Delay in time with the beat.** TIME is set in seconds. Quarter note = 60 / BPM. Eighth note = 30 / BPM. Dotted eighth = 45 / BPM. At 90 BPM, an eighth note is 0.33 s.

MIDI, automation and performance

- **Notes.** MIDNIGHT REELS responds to all MIDI notes (0 to 127) on all MIDI channels, tuned to equal temperament with A4 = 440 Hz. Middle C is MIDI note 60 (C5 in FL Studio).
- **Polyphony.** 16 voices in Poly mode. When you play more than 16 notes, the oldest note is cut to make room; the lowest and highest held notes are protected. In Mono mode, each new note releases the previous one (its release tail still sounds). Releasing a note does not bring back an earlier held note.

- **Velocity.** Velocity changes the volume according to the VEL slider. On the Reel EP engine, velocity also adds bark.
- **Pitch bend.** ±2 semitones. The MIDI pitch wheel and the on-screen wheel add together.
- **Modulation (CC1).** Adds up to 0.8 to the LFO AMOUNT. The higher of MIDI CC1 and the on-screen wheel is used. MIDI CC1 is read by the notes that are sounding when you move the wheel.
- **Sustain pedal (CC64).** Supported. Aftertouch and other MIDI CCs are ignored. Use host automation for everything else.
- **Automation in FL Studio.** Every parameter in the table can be automated. Move a control on the MIDNIGHT REELS panel, then choose **Tools > Last tweaked > Create automation clip**. To use a hardware knob, choose **Tools > Last tweaked > Link to controller** and move the knob. Volume, pitch, bend and cutoff changes are smoothed to avoid zipper noise.
- **Latency.** 0 samples. No delay compensation is needed. The plugin reports a 12-second tail to the host so that delay and reverb tails are not cut when you render.
- **CPU.** MIDNIGHT REELS uses no samples and streams nothing from disk. The CPU load grows with the number of sounding notes. The five-saw Noir Strings engine is the heaviest. Effects at zero are bypassed and use no CPU. Long release times keep voices active longer.
- **Sample rates.** 44.1, 48, 88.2 and 96 kHz.

FAQ

I hear no sound. Check that the MIDNIGHT REELS channel is routed to a Mixer track and that the channel is not muted. Check VOLUME. With Low Pass selected, a very low FILTER setting can silence the sound. With High Pass selected, a high FILTER setting removes bass presets. Finally, remember that the on-screen keyboard plays very low notes in version 1.0: test with a MIDI keyboard or the Piano roll.

MIDNIGHT REELS does not appear in FL Studio. MIDNIGHT REELS is a 64-bit VST3 plugin. Check that `MIDNIGHT_REELS.vst3` is in `C:\Program Files\Common Files\VST3`. In FL Studio, open **Options > Manage plugins** and click **Find installed plugins**. Look for **MIDNIGHT_REELS** under **Generators**, manufacturer **DJBILBOX BEATS**.

Every note has an unwanted vibrato or wobble. Check **AMOUNT** and the on-screen modulation wheel: both add LFO depth. If you use a hardware modulation wheel, return it to zero while you hold a note, so that every voice receives the new position.

The sound gets quieter when I turn REVERB fully down. In version 1.0, the reverb stage also lifts the dry signal by about 6 dB while it is active. At 0 the reverb is bypassed, so the overall level drops. For a dry sound at the usual level, leave REVERB at a very low value such as 0.01, or compensate with VOLUME.

In Mono mode, notes overlap. This is normal. Mono mode releases the previous note instead of cutting it, so its release tail still sounds. Shorten **RELEASE** for a tighter monophonic line.

OCCITANIE

Folk instruments of Southern France: bagpipe, hurdy-gurdy, pipe, mandolin, guitar, accordion, choir and bells, with a built-in drone.

Type	Synthesizer (VST3 instrument)
Format	VST3 64-bit (Windows)
Developer	DJBILBOX BEATS
Version	1.0
Factory presets	70
Latency	0 samples

Overview

OCCITANIE is a 16-voice synthesizer inspired by the traditional instruments of Occitania, the region around Toulouse. It has eight synthesis engines: Cabrette (nasal bagpipe with breath), Hurdy-Gurdy (bowed strings with the typical buzz), Galoubet Pipe (three-hole folk pipe), Mandolin (plucked string with plectrum tremolo), Guitar (plucked string with a soft attack), Accordion (three detuned reeds, musette style), Choir (ensemble with vowel formants) and Carillon (inharmonic bells). It uses no samples. Every note is generated in real time.

A **DRONE** control adds a soft tone one octave below each note, on any engine: the classic bagpipe and hurdy-gurdy sound. Each voice then runs through a multimode filter and a per-voice LFO. The output goes through an effects chain: distortion, chorus, stereo delay and reverb. A soft limiter at the output keeps the level under control.

OCCITANIE brings folk color to modern productions. Use the accordion and pipe sounds for Oriental and Rai melodies, the mandolin and guitar for Afrotrap and Drill hooks, the choirs and bells for cinematic Trap intros, and the hurdy-gurdy and bagpipe for medieval, fantasy and world music. It also works for Deep House toplines and folk-flavored Hip-Hop loops.

Quick start in FL Studio

1. In the **Channel rack**, click **+** and choose **OCCITANIE** (manufacturer **DJBILBOX BEATS**). You can also drag it from **Plugin database > Generators**.
2. The interface opens on **OCCITANIE - Init Patch** (Cabrette engine).
3. Click the **Presets** button at the left of the preset display. Open a bank (for example **Accordion & Musette**) and pick a preset. You can also step through all 70 presets with the **<** and **>** buttons.
4. Play notes from your MIDI keyboard, or draw them in the **Piano roll**.
5. Raise the **DRONE** slider (right) to add a low drone under every note.
6. Move the **REED** slider to change the color of the engine: the nasal tone of the bagpipe, the vowel of the choir, the breath of the pipe.
7. Turn **REVERB** to place the sound in a small room or a large church.
8. Click **Save** at the right of the preset display to save your own version.

Interface

The window opens at 1000 × 746 pixels. Drag the bottom-right corner to resize it from half size (500 × 373) to double size (2000 × 1492). The proportions stay fixed.

Preset display (top center). From left to right: **Presets** opens the preset menu, **<** loads the previous preset, the center shows the current preset name, **>** loads the next preset, and **Save** saves a user preset. The arrows wrap around from the last preset to the first. The preset menu lists the seven factory banks as submenus, a **User** submenu with your saved presets, and an **Open a .occpreset file...** item.

EXPANSION cassette (top left). Click it to see information about upcoming expansion packs (**EXPANSION VOL.1**) and the link to the DJBILBOX BEATS store. Version 1.0 contains no expansion content.

Logo (top right). The DJBILBOX BEATS name and logo.

Medallion (center). The camel warrior, wearing a horned helmet with the Occitan cross, sits between decorative speaker grilles. The OCCITANIE title and “By Humpire” are below it.

LFO group (left). RATE and **AMOUNT** set the LFO speed and depth. **PITCH** sits under the same “LFO” caption but is the global transpose of the instrument, not an LFO control.

Volume envelope (left, below the LFO group). Four vertical sliders: **ATTACK**, **DECAY**, **SUSTAIN** and **RELEASE**.

Timbre sliders and VOLUME (right, below the speaker grille). Five short vertical sliders: **REED**, **BODY**, **BRIGHT**, **DRONE** and **VEL** (velocity sensitivity). The **VOLUME** knob is to their right.

Effects (right). REVERB, **CHORUS** and **DISTORTION** on the upper row. **DELAY** (level), **TIME** and **FEEDBACK** on the lower row.

Center bottom. Two small knobs, **WIDTH** and **TEXTURE**, then the **FILTER** knob (cutoff) and the **GLIDE** knob.

Menu row. Five drop-down menus across the panel. Each one shows its current value. From left to right: the engine (**Waveform**, default “Cabrette”), the LFO shape (**LFO Source**, default “Sine”), the LFO target (**Destination**, default “Pitch”), the filter mode (**Filter Type**, default “Low Pass”) and the voice mode (**Mono/Poly**, default “Poly”).

Wheels (bottom left). The left wheel bends the pitch and springs back to center when you release the mouse. The right wheel is the modulation wheel and stays where you leave it.

On-screen keyboard (bottom). Click a key to audition the sound. It also lights up incoming MIDI notes within its range. In version 1.0 the keyboard covers the lowest part of the MIDI range (keys labeled C-2 to about D1, MIDI notes 0 to 38). Use it to check low notes. Play melodies and chords from a MIDI keyboard or the Piano roll.

Using the controls. Drag a knob up/down or left/right. Drag a slider vertically. The mouse wheel adjusts in small steps. Double-click to return to the default value. Hold Ctrl or Alt while you drag for velocity-sensitive movement. A small bubble shows the current value while you hover or drag.

Parameters

Names in parentheses are the parameter names shown in FL Studio for automation, when they differ from the panel label.

Parameter	Range	Default	Function
Engine menu (Waveform)	Cabrette, Hurdy-Gurdy, Galoubet Pipe, Mandolin, Guitar, Accordion, Choir, Carillon	Cabrette	Selects the synthesis engine. See the engine table below.
LFO shape menu (LFO Source)	Sine, Triangle, Saw, Square, Sample&Hold	Sine	LFO waveform. Sample&Hold picks a new random value once per cycle.

Parameter	Range	Default	Function
LFO target menu (Destination)	Pitch, Filter, Amp, Timbre	Pitch	Where the LFO goes. Pitch: up to ±1 semitone. Filter: up to ±3 octaves of cutoff. Amp: tremolo, up to 50% volume dip. Timbre: moves REED, BODY, BRIGHT, WIDTH and TEXTURE together, up to ±0.5.
Filter mode menu (Filter Type)	Low Pass, High Pass, Band Pass	Low Pass	12 dB/octave state-variable filter mode. Band Pass gets a small level boost.
Voice mode menu (Mono/Poly)	Poly, Mono	Poly	Poly: up to 16 notes. Mono: each new note releases the previous one.
RATE (LFO Rate)	0.05 to 20 Hz	4 Hz	LFO speed. The LFO restarts at each new note. It is not tempo-synced.
AMOUNT (LFO Amount)	0 to 1	0	LFO depth. The modulation wheel (or MIDI CC1) adds up to 0.8 on top.
PITCH (Pitch)	-12 to +12 semitones, continuous	0	Global transpose. Values between semitones detune the whole instrument.
GLIDE (Glide)	0 to 2 s	0 (off)	Portamento. Above 0, each new note slides from the pitch of the previous note, in Poly and Mono.
FILTER (Filter)	40 Hz to 18 kHz	9 kHz	Filter cutoff frequency. The cutoff follows the played note at 25% (key tracking).
REED (Reed / Vowel)	0 to 1	0.5	Engine timbre control 1: reed color, vowel or breath. See the engine table.
BODY (Body / Resonance)	0 to 1	0.5	Engine timbre control 2. See the engine table. BODY also raises the filter resonance slightly.
BRIGHT (Brightness)	0 to 1	0.5	Engine timbre control 3: brightness. See the engine table.
WIDTH (Width / Detune)	0 to 1	0.3	Engine timbre control 4: detune and width. See the engine table.
TEXTURE (Texture)	0 to 1	0.2	Engine timbre control 5: breath, buzz or tremolo. See the engine table.

Parameter	Range	Default	Function
DRONE (Drone)	0 to 1	0	Level of a softened saw one octave below each note. Works on every engine.
VEL (Velocity)	0 to 1	0.5	How much note velocity changes the volume. 0 = every note at full level. 1 = full velocity response.
ATTACK (Attack)	0.001 to 5 s	0.01 s	Volume envelope attack time.
DECAY (Decay)	0.01 to 5 s	0.4 s	Volume envelope decay time.
SUSTAIN (Sustain)	0 to 1	0.7	Volume envelope sustain level.
RELEASE (Release)	0.02 to 8 s	0.4 s	Volume envelope release time after note-off.
VOLUME (Volume)	0 to 1	0.7	Instrument level, before the effects.
DISTORTION (Distortion)	0 to 1	0	Soft saturation. Drive rises up to 15x; the level is compensated.
CHORUS (Chorus)	0 to 1	0	Chorus mix (fixed 0.7 Hz modulation).
DELAY (Delay)	0 to 1	0	Level of the stereo cross-feedback (ping-pong style) delay.
TIME (Delay Time)	0.05 to 1 s	0.35 s	Delay time in seconds. Changes glide smoothly, like a tape delay.
FEEDBACK (Feedback)	0 to 0.9	0.35	Delay repeats. Higher values give more echoes.
REVERB (Reverb)	0 to 1	0.25	Reverb amount. The room also grows larger as you turn it up.
Pitch bend wheel (Pitch Bend)	-1 to +1	0	Bends all notes by up to ±2 semitones. Adds to the MIDI pitch wheel.
Modulation wheel (Mod Wheel)	0 to 1	0	Adds LFO depth (up to +0.8). The higher of this wheel and MIDI CC1 is used.

Effects order: Distortion, Chorus, Delay, Reverb, then the output soft limiter. Each effect is bypassed while its knob is at zero. OCCITANIE has no filter envelope: shape the sound over time with the volume envelope and the LFO.

What the timbre controls do on each engine

Engine	REED	BODY	BRIGHT	WIDTH	TEXTURE
Cabrette: two saws with a nasal double formant and an attack puff	Formant position (nasal color)	Formant level	Brightness	Detune between the two saws	Breath noise and attack puff
Hurdy-Gurdy: two saws with formants and a buzzing bridge	Formant position	Formant level	Brightness	Detune between the two saws	Buzz (bridge saturation)
Galoubet Pipe: soft harmonic tone with breath	Breath amount (more air, less tone)	Breath focus (resonance of the air noise)	Upper harmonics	Second, detuned pipe layer	Chiff (attack noise)
Mandolin: plucked string model with plectrum tremolo	No effect	Sustain	Brightness	No effect	Plectrum tremolo depth (14 Hz)
Guitar: plucked string model with a soft attack	No effect	Sustain (long at high values)	Brightness	No effect	No effect
Accordion: three detuned reeds with formants	Formant position	Formant level	Brightness	Musette detune between the reeds	Bellows noise
Choir: five detuned saws with vowel formants	Vowel (formant position)	Formant level	Brightness	Ensemble detune	Breath noise
Carillon: four inharmonic bell partials	No effect	Ring length	Upper partials	No effect	Strike noise

On every engine, BODY also controls the filter resonance (from flat at 0 to a gentle peak at 1).

Factory presets

OCCITANIE includes 70 factory presets in 7 banks of 10. Loading a factory preset resets every parameter before applying the preset. The pitch bend and modulation wheels are not changed. Most banks use the engine of the same name; exceptions are noted. Place names of the region (Capitole, Canal du Midi, Garonne, Albi, Carcassonne…) are kept as they are.

Cabrette & Bagpipes (Cabrette engine) Capitole Cabrette, Canal du Midi, Garonne Drone, Pic du Midi, Village Festival, Sad Cabrette, Folk Dance Bagpipe, Corbieres, Blue Woad, Saint Sernin.

Hurdy-Gurdy & Bowed Strings (Hurdy-Gurdy engine) Wheel of Strings, Trumpet String Buzz, Albi Drone, Minstrel, Troubadour, Cathar Chant, Rouergue, Tarn Mist, Melody String, Cordes sur Ciel.

Flutes & Galoubet (Galoubet Pipe engine) Joyful Galoubet, Tabor Air, Shepherd’s Flute, Camargue Fife, Pastoral, Mistral, Tramontane, Cathar Flute, Occitan Whistle, Corbieres Dawn.

Plucked Strings (Mandolin engine: Toulouse Mandolin, Mandolin Tremolo, Cittern, Soft Plectrum; Guitar engine: the other six) Toulouse Mandolin, Mandolin Tremolo, Gypsy Guitar, Medieval Lute, Cittern, Soft Nylon Guitar, Soft Plectrum, Southern Strum, Minstrel Pizz, Low Theorbo.

Accordion & Musette (Accordion engine) Toulouse Musette, Musette Dance Hall, Violet Waltz, Cabaret Accordion, Southern Java, Slow Bellows, Chapel Harmonium, Barrel Organ, Gavotte, Cassoulet Musette.

Choirs & Pads (Choir engine; Ariege Mist uses Hurdy-Gurdy) Cathar Choir, Montsegur, Carcassonne Pad, Medieval City, Voice of Oc, Albi Plainchant, Moon of Oc, Moissac Abbey, Ariege Mist, Saint Sernin Crypt.

Carillons & Textures (Carillon engine; Autan Wind uses Galoubet Pipe, Cevennes Storm uses Hurdy-Gurdy) Capitole Carillon, Saint Etienne Bells, Evening Angelus, Albi Bell Tower, Pink City Glockenspiel, Ariege Crystals, Garonne Dew, Autan Wind, Cevennes Storm, Stars of Oc.

User presets. Save writes the current sound as a `.occpreset` file in `Documents\DJBILBOX BEATS\OCCITANIE\Presets`. Files in this folder appear in the **User** submenu of the preset menu. Use **Open a .occpreset file...** to load a preset from another folder. Your sound is also saved inside your FL Studio project.

Production tips

- **Bagpipe lines.** Start from **Canal du Midi**: Cabrette engine, Mono, short glide and DRONE at 0.5. **REED** moves the nasal formant, **TEXTURE** adds breath and a puff at each attack. The drone is part of each voice: it follows the note you play and stops with it. For a fixed drone under a moving melody, hold one long low note on a second OCCITANIE channel.
- **Oriental and Rai melodies.** Use the Accordion engine with **WIDTH** between 0.6 and 0.85 for a wide musette detune. Switch to **Mono** and set **GLIDE** to 0.05 to 0.1 s for slides between notes. Use the pitch wheel (±2 semitones) for ornaments. For vibrato, set the LFO to **Pitch**, **RATE** to 5 or 6 Hz and push the modulation wheel only on long notes.
- **Hurdy-gurdy buzz.** On the Hurdy-Gurdy engine, **TEXTURE** drives the buzzing bridge. **Trumpet String Buzz** pushes it to 0.85. Combine it with **DRONE** and a slow **ATTACK** (0.1 to 0.3 s) for a bowed, medieval feel.
- **Plucked hooks for Afrotrap and Drill. Toulouse Mandolin** and **Southern Strum** cut through a busy beat. On the Mandolin engine, **TEXTURE** adds the fast plectrum tremolo. On both string engines, **BODY** sets how long the string rings and **BRIGHT** how bright it sounds. **Low Theorbo** gives a deep plucked bass line in Mono.
- **Choirs and chant.** On the Choir engine, **REED** selects the vowel. **Voice of Oc** sends a slow LFO to **Timbre** so the vowel keeps moving. Add **DRONE** (0.3 to 0.4) for a plainchant effect, as in **Albi Plainchant**.
- **Bells for intros.** On the Carillon engine, **BODY** sets the ring length and **BRIGHT** the upper partials. Set **PITCH** to -12 for large tower bells (**Saint Etienne Bells**). A dotted delay and a long reverb turn a single hit into an intro.
- **Layering in a Trap or Hip-Hop mix.** Folk sounds can fill the low mids. Choose **High Pass** in the filter mode menu and raise **FILTER** to 150 to 300 Hz to make room for the 808 and the kick. Lower **DRONE** on dense chords.
- **Delay in time with the beat.** TIME is set in seconds. Quarter note = 60 / BPM. Eighth note = 30 / BPM. Dotted eighth = 45 / BPM. At 100 BPM, a quarter note is 0.6 s.

MIDI, automation and performance

- **Notes.** OCCITANIE responds to all MIDI notes (0 to 127) on all MIDI channels, tuned to equal temperament with A4 = 440 Hz. Middle C is MIDI note 60 (C5 in FL Studio).
- **Polyphony.** 16 voices in Poly mode. When you play more than 16 notes, the oldest note is cut to make room; the lowest and highest held notes are protected. In Mono mode, each new note releases the previous one (its release tail still sounds). Releasing a note does not bring back an earlier held note.
- **Velocity.** Velocity changes the volume according to the VEL slider.
- **Pitch bend.** ±2 semitones. The MIDI pitch wheel and the on-screen wheel add together.
- **Modulation (CC1).** Adds up to 0.8 to the LFO AMOUNT. The higher of MIDI CC1 and the on-screen wheel is used. MIDI CC1 is read by the notes that are sounding when you move the wheel.
- **Sustain pedal (CC64).** Supported. Aftertouch and other MIDI CCs are ignored. Use host automation for everything else.

- **Automation in FL Studio.** Every parameter in the table can be automated. Move a control on the OCCITANIE panel, then choose **Tools > Last tweaked > Create automation clip**. To use a hardware knob, choose **Tools > Last tweaked > Link to controller** and move the knob. Volume, pitch, bend and cutoff changes are smoothed to avoid zipper noise.
- **Latency.** 0 samples. No delay compensation is needed. The plugin reports a 12-second tail to the host so that delay and reverb tails are not cut when you render.
- **CPU.** OCCITANIE uses no samples and streams nothing from disk. The CPU load grows with the number of sounding notes. The five-saw Choir engine is the heaviest. Effects at zero are bypassed and use no CPU. Long release times, common on pads and bells, keep voices active longer.
- **Sample rates.** 44.1, 48, 88.2 and 96 kHz.

FAQ

I hear no sound. Check that the OCCITANIE channel is routed to a Mixer track and that the channel is not muted. Check VOLUME. With Low Pass selected, a very low FILTER setting can silence the sound. With High Pass selected, a high FILTER setting removes low notes. Finally, remember that the on-screen keyboard plays very low notes in version 1.0: test with a MIDI keyboard or the Piano roll.

OCCITANIE does not appear in FL Studio. OCCITANIE is a 64-bit VST3 plugin. Check that `OCCITANIE.vst3` is in `C:\Program Files\Common Files\VST3`. In FL Studio, open **Options > Manage plugins** and click **Find installed plugins**. Look for **OCCITANIE** under **Generators**, manufacturer **DJBILBOX BEATS**.

The drone changes pitch with my melody. This is by design. DRONE adds a tone one octave below each note you play, so it moves with the melody. For a fixed drone, play one sustained low note on a separate channel.

Where is the filter resonance control? OCCITANIE has no separate resonance knob. BODY raises the resonance gently while it shapes the engine.

The sound gets quieter when I turn REVERB fully down. In version 1.0, the reverb stage also lifts the dry signal by about 6 dB while it is active. At 0 the reverb is bypassed, so the overall level drops. For a dry sound at the usual level, leave REVERB at a very low value such as 0.01, or compensate with VOLUME.